MidiPad User Manual

version 0.238

1. Welcome

MidiPad turns a game controller into a MIDI instrument and, above all, a MIDI looper: sticks, triggers and the touchpad send continuous controllers (CC), buttons play notes and chords, and every channel can carry its own layout. It runs as a standalone program next to your DAW or synthesizer and sends MIDI to any port on your system.

MidiPad was designed around the Sony DualSense (PlayStation 5) controller, whose touchpad tracks two fingers. Other controllers recognised by SDL3 work too, without the touchpad features.

A MIDI looper in your hands

Looping is what MidiPad does best. Hold one button, play, release: what you just played repeats at once, in time with itself, and your hands are free again to play over it. No DAW recording, no clip to set up, no bar count to choose beforehand.

- **Loop notes and chords:** a chord progression, a bass line or a riff played on the buttons, the D-pad or the touchpad pads loops with its exact timing and velocity.
- **Loop movements:** a stick gesture, a trigger squeeze or a finger slide on the touchpad loops as a CC curve, a hand-drawn LFO for a filter, a pan or an effect send.
- **Layer a whole arrangement:** each of the sixteen MIDI channels has its own loops, and all of them play together. Loop chords on channel 1, switch to channel 2 and loop a bass line, then add a filter sweep on top.
- **Keep playing live:** movement loops play on CCs of their own, beside the control that recorded them, so the stick or finger stays free to play live over its own loop.
- **Shape it as you go:** add notes with overdub, play loops forwards, backwards or back and forth, pause and restart every loop at once with Play and Stop, and, with Shift, stretch a movement loop up to 100 times longer or 20 times shorter, or narrow its swing, while it plays, and lock a loop so that Record keeps it.

Because the loops are plain MIDI, they drive any instrument: a plugin in your DAW, a standalone synthesizer or a hardware synth on a MIDI interface. Chapter 9 explains looping in detail.

What you can do with MidiPad

- **Play notes and chords** from the face buttons, the D-pad, the shoulder buttons and the four zones of the touchpad, momentarily or latched (toggle).
- **Sculpt sounds** with the two sticks, the analog triggers L2/R2 and two touchpad fingers, each sending a CC with its own range and direction.
- **Freeze** a stick at its current value, so a filter or effect stays put while your hands do something else.
- **Record loops** of notes and movements on every channel, and layer them live (see above); lock the ones to keep, and keep up to eight clips per channel, launched in time from the session screen.

- **Play in time** with MidiPad's own clock, or with your DAW through Ableton Link: loops snap to whole beats or bars, their notes onto a grid if you like, and follow the tempo, which you can set, type or tap. Or play free.
- **Edit loop notes** in a large piano roll: move a note, a chord or a selection (onto the scale if you like), change their length, or delete them, and shorten or lengthen the loop, without recording again.
- **Save sessions**: every channel's clips, the preset, the tonality and the tempo in one file, to open again later.
- **Switch MIDI channels** on the fly, each channel with its own mapping, so one controller drives a whole multitimbral setup.
- Give every button a second meaning with a Shift layer: while a Shift button is held or latched, the buttons, the D-pad and the touchpad pads play their Shift mapping.
- **Change programs** on your synthesizer from a button or by sweeping a stick through a range of presets.
- **Humanize** velocity and timing for a less mechanical feel.

Who this manual is for

This manual assumes you know what MIDI notes, channels and CCs are, and that you have a DAW or synthesizer on your computer. It does not assume you have used MidiPad before: chapters 2 and 3 take you from installation to your first sound, and the later chapters describe every part of the window in order.

The manual describes MidiPad on **Linux**. Notes for Windows and macOS will appear in boxes marked *Windows* and *macOS* when those versions are available.

Conventions

Notation	Meaning
Options	A label, button or menu entry as it appears on screen
☰ › **Options**	A menu path: open the ☰ menu, then choose Options
`Ctrl+S`	A keyboard shortcut
Cross, Circle, Square, Triangle	The DualSense face buttons (A, B, X, Y on other controllers)
L1/R1, L2/R2, L3/R3	Shoulder buttons, analog triggers, stick clicks
CC 40	MIDI Control Change number 40
C4 = 60	Note names: middle C is C4, MIDI note 60

> **Note:** boxes like this one give tips and details you can skip on a first reading.

2. Installation on Linux

MidiPad is built from source with CMake; the build downloads its own libraries, so you only need a compiler and the ALSA headers. Installation takes about five minutes on a recent machine.

Requirements

Item	Version or package
Compiler	C++20: GCC 13 or newer (tested), or a recent Clang
Build tools	CMake 3.25 or newer, Ninja, Git
ALSA headers	`libasound2-dev` (Debian, Ubuntu), `alsa-lib-devel` (Fedora)
JACK	Optional
Controller	DualSense over USB or Bluetooth; any SDL3 gamepad without touchpad

On Debian or Ubuntu, install everything in one command:

```
sudo apt install build-essential cmake ninja-build git libasound2-dev
```

Building

1. Get the source code and enter its folder.
2. Configure, build and (optionally) run the tests:

```
cmake --preset release
cmake --build --preset release
ctest --preset release
```

The program is created at `build/release/src/app/midipad`. You can start it from there, or copy it to a folder on your `PATH`, such as `~/.local/bin`.

Giving MidiPad access to the touchpad

The sticks, triggers and buttons work as soon as the controller is connected. The DualSense touchpad is read directly from the device (`/dev/hidraw*`), which needs a udev rule. Without it, MidiPad works but the TOUCHPAD module stays inactive, and MidiPad tells you so when the controller connects.

Create the file `/etc/udev/rules.d/70-dualsense.rules` with this content:

```
# USB
KERNEL=="hidraw*", SUBSYSTEM=="hidraw", ATTRS{idVendor}=="054c",
ATTRS{idProduct}=="0ce6", MODE="0666"
# Bluetooth
KERNEL=="hidraw*", SUBSYSTEM=="hidraw", KERNELS=="0005:054C:0CE6.*", MODE="0666"
```

Then reload the rules and reconnect the controller:

```
sudo udevadm control --reload-rules && sudo udevadm trigger
```

Connecting the controller

- **USB:** plug in the cable. The controller is ready at once.
- **Bluetooth:** hold **Create** and the **PS** button until the light bar flashes, then pair *DualSense Wireless Controller* in your system's Bluetooth settings.

MidiPad finds the controller whether you connect it before or after starting the program. The status bar at the bottom right of the window shows its name and a green dot when it is connected.

First launch

Start MidiPad:

```
build/release/src/app/midipad
```

On the very first launch, MidiPad does not know where to send its MIDI yet, so the **MIDI output** window asks you to choose a port. Chapter 3 explains which one to pick. Until you choose, MidiPad sends to its own virtual port, named *MidiPad*.

> **Note:** MidiPad keeps reading the controller while its window is in the background, so you can play while your DAW has the focus.

> **Windows / macOS:** installation notes will be added with those versions.

3. Quick start

Five steps take you from a fresh installation to playing a synthesizer in your DAW: connect, choose an output, set up a track, play, save.

Step 1: connect the controller and start MidiPad

Connect the DualSense by USB or Bluetooth, then start MidiPad. The green dot at the bottom right confirms that the controller is seen.

Step 2: choose where the MIDI goes

On the first launch, the **MIDI output** window lists every MIDI port on your system.



Pick the port your DAW can read:

- **Virtual port "MidiPad"**: a software port. DAWs that use the ALSA sequencer or JACK (Reaper, Ardour, Qtractor) see it directly as an input named *MidiPad*.
- **VirMIDI ports**: kernel MIDI ports, seen by every program, including those that only list hardware devices. Choose **VirMIDI 0-0**, then select the input shown on the right (*Virtual Raw MIDI*) in your DAW. If no VirMIDI port exists, the window offers to create them (see chapter 11).
- **Later**: keep the virtual port for now; the question comes back at the next start.

MidiPad remembers your choice and reconnects to the same port at every start. You can change it at any time in ☰ › **Options**.

> **Tip:** if you are unsure, start with VirMIDI 0-0. It works with every DAW on Linux.

Step 3: set up a track in your DAW

1. Create an instrument track with a synthesizer.
2. Set the track's MIDI input to the port chosen in step 2, on all channels or channel 1.
3. Arm or monitor the track so it plays what it receives.

Step 4: play

- Press **Cross**: the chord shown under the Cross pad sounds. The pad lights up while the chord plays.
- Move the **left stick**: it sends CC 40 (left-right) and CC 41 (up-down). Use your DAW's MIDI learn to assign them to a filter cutoff or any parameter.
- Press a **D-pad** direction: in the demo preset, Up, Right, Down-left and Left latch chords (press again to stop them).

Every change you hear is also shown in the window: the stick's screen follows your thumb, the CC pills light up, and the piano roll scrolls the notes you play.

Step 5: make it yours and save

Click any control in the window to change what it sends: a pad, a D-pad row, a CC pill, a module's screen. Chapter 6 describes these mapping windows. When you are happy, press `Ctrl+S` or click **Save preset**.

> **Note:** changes are heard at once but written to disk only when you save. Unsaved changes are kept if you only switch channels; MidiPad asks before discarding them when you open another preset or quit.

4. The main window

MidiPad has a single window that plays and edits at the same time: every module mirrors a part of the controller, shows what it sends, and opens its settings when you click it. The window is built for the stage, so everything you need while playing is one click away.

The screenshots in this manual use a preset called *Electronica Demo*, in F minor, with loops playing.



#	Area	What it does
1	☰ menu	Open, Save, Save as (sessions), Options, Fullscreen (`F11`), About, Quit (`Ctrl+Q`)
2	Channel selector	The active MIDI channel (1–16); arrows step through channels
3	Preset menu	Opens, creates, saves and deletes presets; arrows load the previous or next preset
4	Play / Stop	Restarts or pauses the loops of every channel; Play turns green while loops run. With Ableton Link, they can also start and stop the Link session

#	Area	What it does
5	Clock (LINK / CLOCK)	FREE without a clock; otherwise the tempo of the internal clock or of the Link session (with the number of Link peers) and the beats of the bar. The wheel or a double-click on the tempo changes it; a click elsewhere turns the clock off or back on
6	TAP	Tap tempo: tap in time and the clock's tempo follows, from the second tap
7	Session	Shows the session screen, the clips of every channel, in place of the modules (`Tab`); lit green while it shows
8	PANIC	Silences everything at once: all notes off, loops stopped, delayed notes dropped
9	LEFT STICK	Stick position, its CCs, the stick-click (L3) action and its loop
10	RIGHT STICK	The same for the right stick (R3)
11	TRIGGERS	L2 and R2 travel, their CCs, their button actions and their loop
12	D-PAD	The eight directions, including diagonals, and what each plays
13	TOUCHPAD	Finger positions and their CCs, or the four pad zones, and the finger loops
14	Piano roll	The notes of the active channel, scrolling or as a loop clip
15	SCALE	The tonality; click to choose a root and a scale
16	CHANNEL	The active channel in large digits and the sixteen channels; a violet dot marks channels with loops
17	Button strip	One pad per button: L1, L2, L3, Create, Cross, Circle, Square, Triangle, Options, PS, Touchpad click, Mute, R3, R2, R1
18	MIDI out	The port MidiPad sends to, and the name your DAW sees it under
19	Controller	The connected controller, and *+ touchpad* when the touchpad is readable

Reading the window

- **A green dot** in a module's header means the controller is connected and that part is active.
- **Pills** such as *X CC 40* show what an axis sends; they light up as the value moves. *inv* marks an inverted axis, *PC 1-16* an axis that selects programs.
- **Pads** in the button strip flash on every press and stay lit while their note or chord sounds. A latched (toggle) chord stays lit until released. The label under each pad shows its action: a chord name, a note, *Freeze*, *Ch +1*, *Rec*, *Shift*, *PC 1* and so on; - means nothing is assigned. While Shift is on, the pads, the D-pad rows and the touchpad zones show their Shift layer.
- **The ↻ icon** next to a D-pad chord marks a toggle: press once to start, again to stop.

Clicking to edit

Click a control to open its mapping window over the view: a pad, a D-pad row, a CC pill, a module's screen, a LOOP panel or the piano roll. On channel 1, the window edits channel 1's mapping, which every channel without its own plays; on another channel, your first change gives that channel its own mapping (chapter 8). The view keeps playing behind it, so you hear each change at once.

From the controller: edit mode, menus and help

MidiPad can be set up with the controller in your hands. Playing never changes a setting by accident: the controller drives the interface only in the session screen (chapter 9) and in the **edit mode**, which you open on purpose.

The buttons. The **Mode** action, given to a button in the mapping (chapter 6; PS / Guide without one), opens and closes these screens:

- **Mode:** the session screen, and back.
- **Shift + Mode:** the edit mode, and back.
- **Mode held** half a second: the help of the screen you are on, with your own buttons; it goes when you let go.

Three more buttons drive these screens, chosen in ☰ › Options › CONTROLLER (chapter 11): **Select** (launch, choose, open; L3 and R3 when none is chosen), **Menu** (the menu of what is under the cursor) and **Back** (close a menu, leave the edit mode). There, they do this instead of their usual action; on the main screen they play as usual, so a button can play a chord and be Select in the session screen.

The edit mode. The session button of the top bar turns violet and reads **EDIT**, and a cyan cursor goes over the main screen: the lanes of the LOOP panels, the piano roll, the screens of the modules, their pills and rows, the pads.

- The left stick or the D-pad moves the cursor to the nearest item that way.
- **Select** opens what is under it, as a click: the window of a control, the note loop editor. In a window, the D-pad goes from field to field and Select changes the field; **Back** cancels a field being typed, then closes the window.
- **Menu** opens the menu of the item (below). **Back**, outside a window, leaves the edit mode, as Shift + Mode does.

The buttons that are not Select, Menu or Back still play.

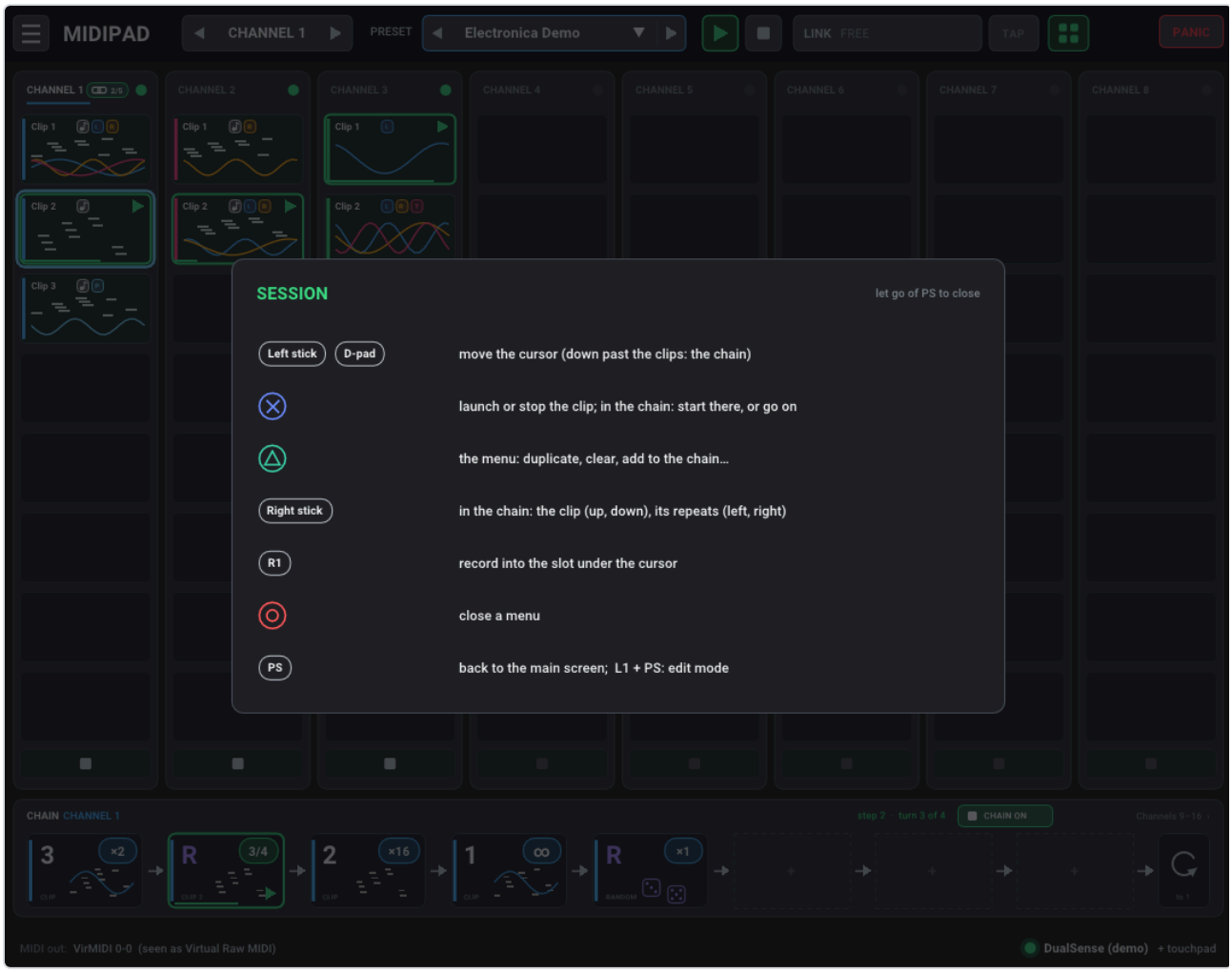
Menus. A right-click, or Menu with the controller, opens a small menu beside the item, with what can be done to it:

- **A lane** of a LOOP panel: Lock or Unlock, Clear that axis, Back to 1x · 100%, Settings...; **the header of a panel**: the same for all of its axes.
- **The piano roll**: Lock or Unlock the note loop, Edit the notes..., Clear.

- **A clip** in the session screen: Launch or Stop, Record here, Duplicate (into the first free slot), Add to the chain, Clear.
- **A step of a chain**: Start from here, Random (R), Insert a step, Delete the step, Empty it.
- **A channel** (the header of its column, or the end of its chain): Stop, Chain on or off, End: stop or back to step 1, Clear the chain.

With the controller, the D-pad or a stick goes up and down the menu (greyed items are skipped), Select chooses, Menu or Back closes it.




MIDIPAD
CHANNEL 1
PRESET
Electronica Demo
LINK FREE
TAP
PANIC
SESSION
let go of PS to close
Left stick
D-pad
move the cursor (down past the clips: the chain)
launch or stop the clip; in the chain: start there, or go on
the menu: duplicate, clear, add to the chain…
Right stick
in the chain: the clip (up, down), its repeats (left, right)
R1
record into the slot under the cursor
close a menu
PS
back to the main screen; L1 + PS: edit mode
CHAIN CHANNEL 1
CHAIN ON
MIDI out: VirMIDI 0-0 (seen as Virtual Raw MIDI)
DualSense (demo) + touchpad

5. Control modules

Each module stands for one part of the controller and has the same layout from top to bottom: an animated screen that follows your hands, a MIDI area showing what is sent, and a LOOP panel for the recorded movements.

Sticks

Each stick sends two CCs, one per axis. In the demo preset the left stick sends CC 40 (X) and CC 41 (Y), the right stick CC 42 and CC 43, with Y inverted so that pushing up raises the value. A new preset starts empty: every axis and button is unassigned until you set it.

- **Screen:** a glowing shape follows the stick. Its size grows with the distance from the centre.
- **Pills:** *X CC 40* and *Y CC 41 inv* show the CC numbers; *inv* marks an inverted axis.
- **Stick click row:** shows the action of L3 or R3, with *free* or *frozen* on the right.

A stick sends a value only when it changes, so a stick at rest adds no traffic.

Freeze

A button set to **Axis freeze** (L3 and R3 in the demo) holds the chosen axes at their current value: one press freezes, the next one releases. While frozen, the shape turns ice blue and keeps following your thumb, a ghost and a crosshair mark the frozen position, and the screen shows **FROZEN**. Bring the stick back to the crosshair before releasing if you want the value to continue smoothly.

Triggers

L2 and R2 are both analog axes and buttons. As axes they send a CC over their whole travel (CC 50 and CC 51 in the demo). As buttons they can play a note, a chord or any other action: they press at 25 % of the travel and release at 15 %, so a light touch does not trigger them by accident.

The TRIGGERS module shows both travels side by side, the CC pill and the button action of each trigger, and a LOOP panel with both recorded curves.

D-pad

The D-pad plays eight directions: the four arrows and the four diagonals. Each row of the module shows a direction, its action and, for a toggle, the ↻ icon. The screen shows a 3D D-pad with the pressed direction raised in red, and the name of what it plays.

Your thumb never presses or releases the two arrows of a diagonal at exactly the same moment. To keep the arrow crossed on the way silent, a new direction plays only after it has stayed stable for a short time: 15 ms by default, adjustable in Options. Releasing the whole D-pad stops the sound at once.

Touchpad

The DualSense touchpad tracks two fingers. Each finger works in one of three modes, chosen in the touchpad window:

Mode	What the finger does
Off	Nothing
X/Y position → CC	Sends two CCs: left-right and bottom-top (finger 1: CC 60/61, finger 2: CC 62/63 in the demo)
4 zones → pads	Splits the touchpad into four quarters; touching a quarter plays its note or chord

The screen shows each finger as a numbered dot. In pad mode it shows the four zones and lights the one being touched.

> **Note:** the touchpad needs the udev rule described in chapter 2. Without it, the module stays inactive and the status bar does not show *+ touchpad*.

Button strip

The strip at the bottom has one pad per button, in this order: L1, L2, L3, Create, Cross, Circle, Square, Triangle, Options, PS, Touchpad click, Mute, R3, R2, R1. The face buttons use their PlayStation colours. Click a pad to choose its action.

Piano roll, SCALE and CHANNEL

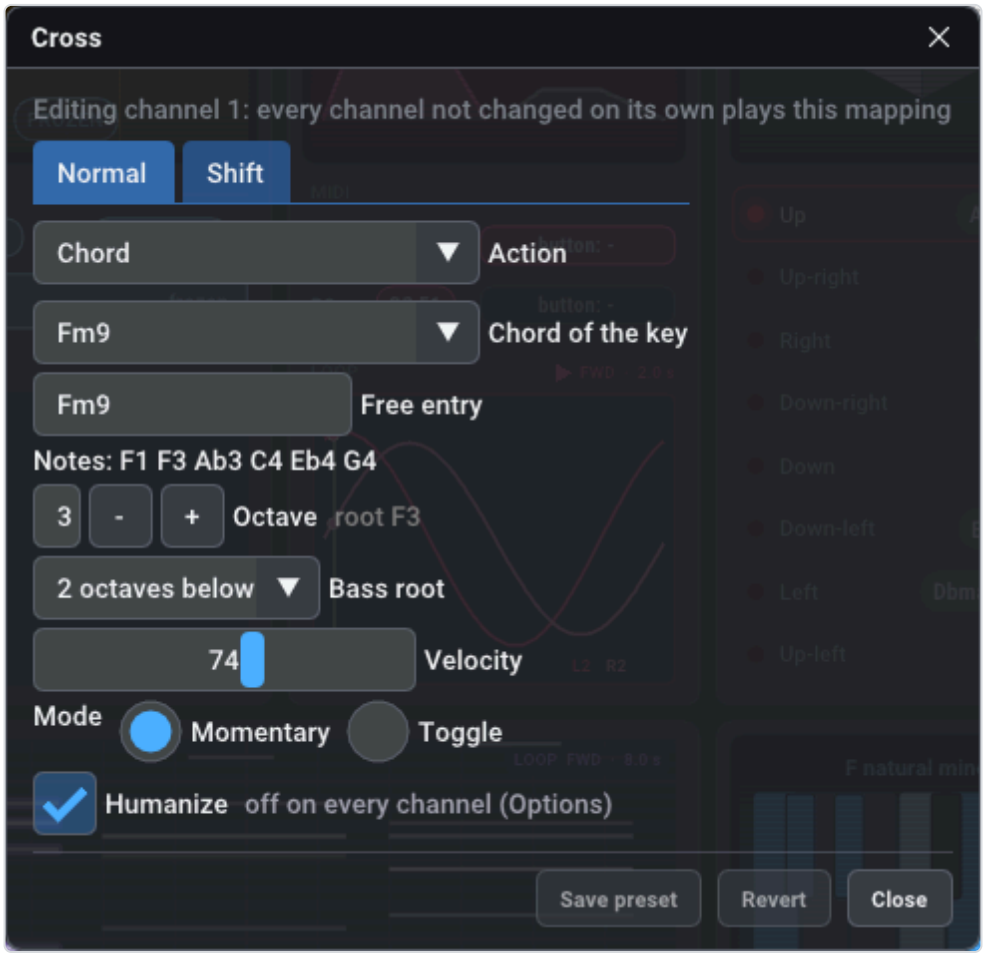
Below the modules, three blocks show the musical state of the active channel:

- **Piano roll:** the notes and chords played, scrolling towards the *now* line on the right; when a note loop runs, it turns into a clip of the whole loop; click it to open the note loop editor (chapter 9).
- **SCALE:** the chosen tonality on a one-octave keyboard, with the notes of the scale lit. *no scale* means none is chosen. Click it to choose one (chapter 7).
- **CHANNEL:** the active channel in large digits, and a grid of the sixteen channels. A violet dot marks every channel where a loop is playing.

6. Mapping windows

A mapping window sets what one control sends. Click a control in the main window to open it; the window floats over the view, which keeps playing, so you hear every change as you make it.

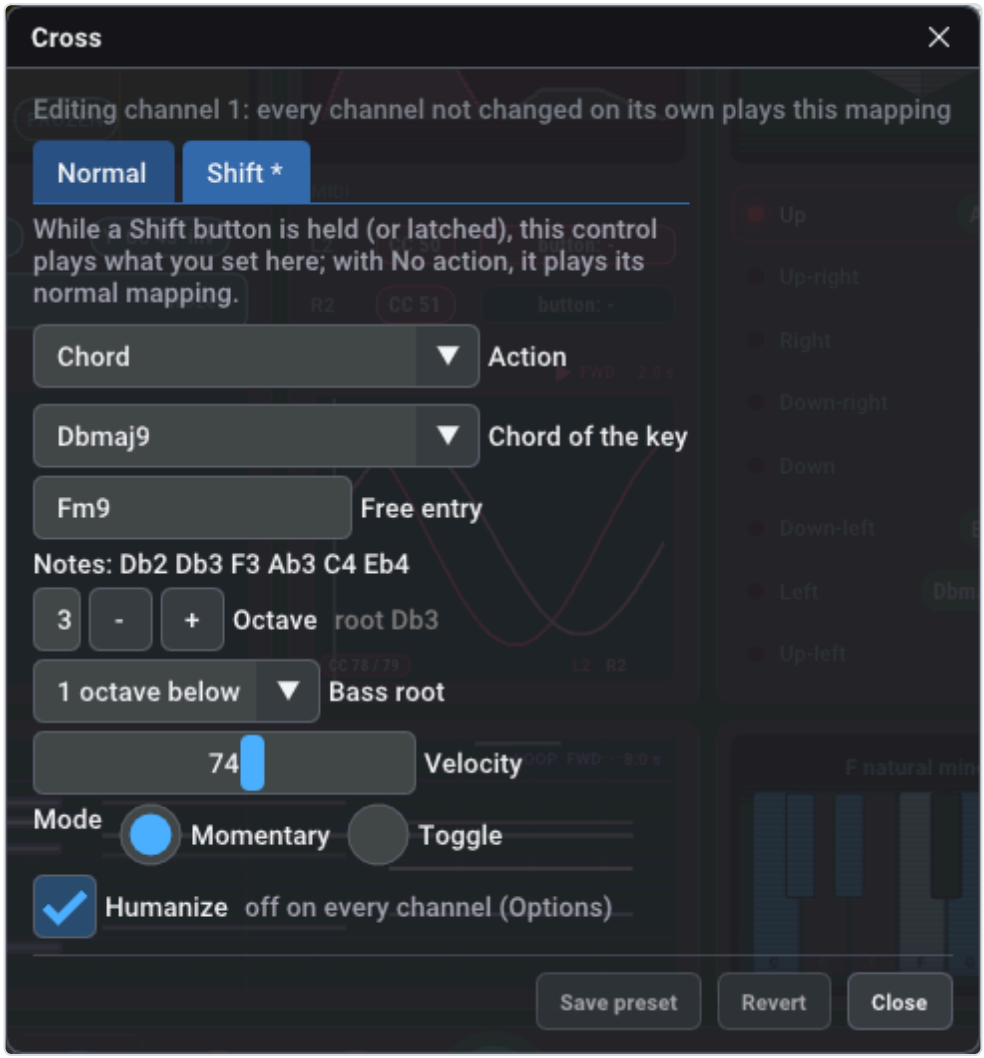
Anatomy of a mapping window



- **Title:** the control being edited (here *Cross*).
- **Scope line:** on channel 1, *Editing channel 1: every channel not changed on its own plays this mapping*; on another channel, *...it plays channel 1's mapping until you change it*, then *Editing the mapping of channel N (its own)*, with a **Use channel 1's mapping again** button (chapter 8). The note loop editor has none: a loop belongs to its channel.
- **Normal and Shift tabs** (buttons, D-pad directions and touchpad pads): what the control plays normally, and what it plays while Shift is on. The tab reads *Shift ** once something is set there.
- **Action:** the type of action; the fields below change with it.
- **Save preset** (`Ctrl+S`) writes the preset to disk. **Revert** reloads the saved version and discards your changes. **Close** (or `Esc`) closes the window and keeps your changes in memory.

The Shift layer

Every button, D-pad direction and touchpad pad has two layers. The **Normal** tab sets what it plays; the **Shift** tab sets what it plays while a Shift button is held, or latched in Toggle mode. A control whose Shift tab stays on *No action* plays its normal mapping under Shift too, so only the controls you set there change.



Give a button or a D-pad direction the **Shift** action to reach the layer (L1 in the demo); a Shift control has no Shift layer of its own. Each channel's mapping has its own layer. While Shift is on, the main window shows the layer on the pads, the D-pad and the touchpad, and the sticks and fingers reshape their loops instead of sending CCs (chapter 9).

A window opens on the layer shown last: after working on the Shift tab of one control, the next control you click opens on its Shift tab too. A control clicked while Shift is on opens on its Shift tab.

Button actions

Every button, D-pad direction, trigger (as a button) and touchpad zone can take one of these actions:

Action	Sends or does	Main settings
No action	Nothing	—
Note	One note	Note, Velocity, Mode, Humanize
CC	A Control Change	CC number, Value on press, Value on release, Mode
Chord	Several notes at once	Chord, Octave, Bass root, Velocity, Mode, Humanize
Transport	MIDI Start, Stop or Continue	Command
Channel change	Moves to the previous or next channel	Previous (-1) or Next (+1)
Axis freeze	Holds axes at their current value	Axes to freeze
Loop record	Records loops while held	— (see chapter 9)
Program change	Selects a program on the synthesizer	Program, Next or Previous, Bank select (chapter 10)
Shift	While on, the buttons, D-pad and touchpad pads play their Shift layer, and sticks and fingers reshape their loops	Mode (see The Shift layer above, and chapter 9)
Tap tempo	Sets the clock's tempo from your taps, from the second one	— (see chapter 9)
Mode	A press opens or closes the session screen; with Shift, the edit mode; held, the help (chapter 4)	— (see chapter 9); without one in the preset, PS / Guide does it

Mode: Momentary or Toggle

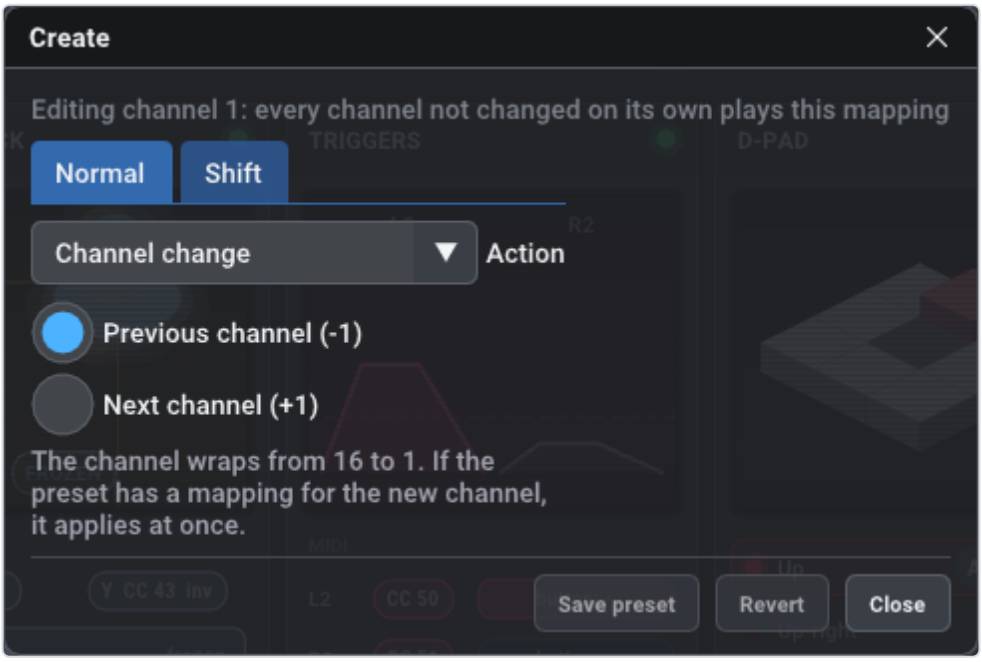
- **Momentary:** the note, chord or CC lasts as long as you hold the control. A CC sends *Value on press*, then *Value on release*.
- **Toggle:** the first press starts, the second stops. A latched note or chord keeps sounding with your hands free.

Toggles within one family of controls exclude each other: starting a latched chord on the D-pad stops the one already latched there. The families are the buttons, the D-pad, and each touchpad finger. CC toggles are always independent.

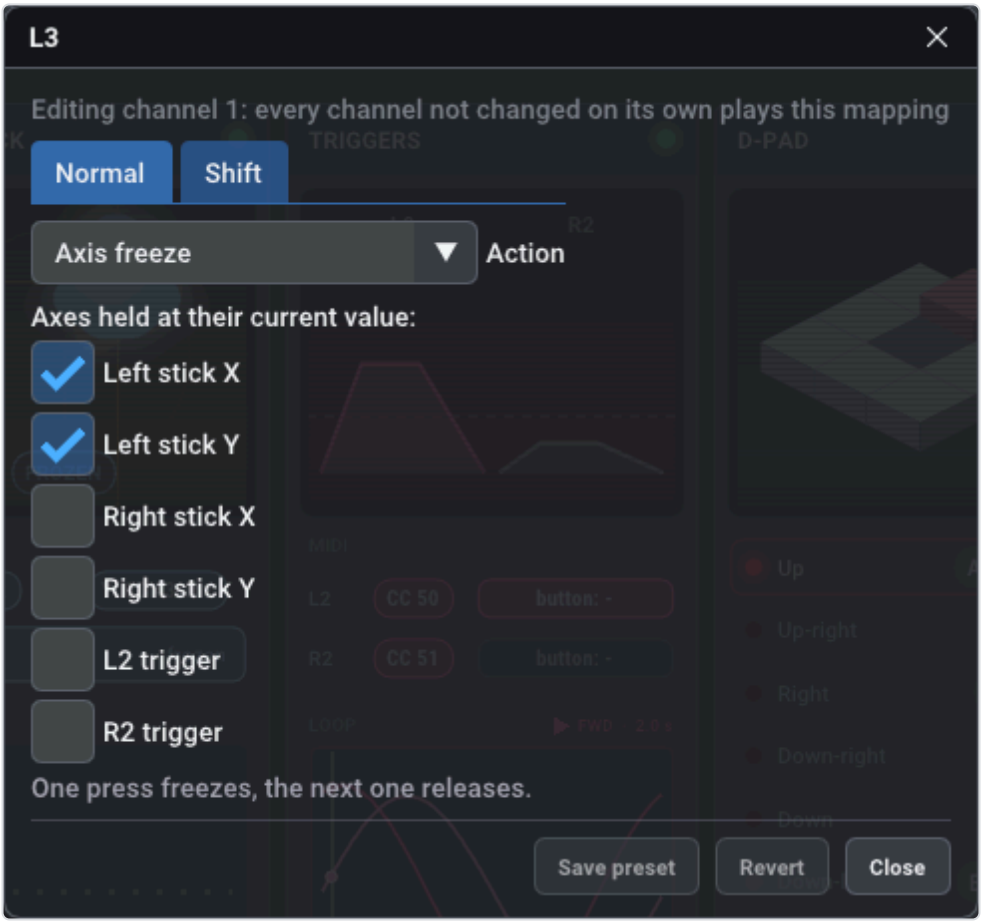
Velocity and Humanize

Velocity (1–127) sets how hard the note is played. The **Humanize** box lets this control follow the random velocity and timing set for its channel in Options (chapter 11); the text beside it tells you whether humanization is active on any channel. Untick it to keep this control exact, for a kick drum for example.

Other button windows



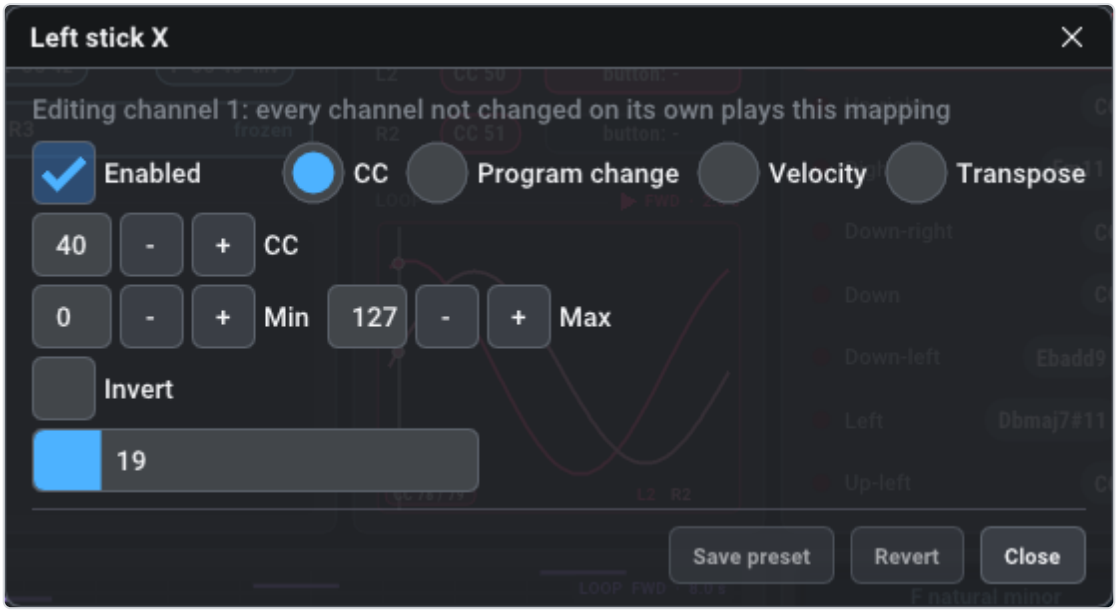
Channel change steps through the sixteen channels, wrapping from 16 to 1. If the preset has a mapping for the new channel, it applies at once.



Axis freeze holds any combination of the six axes: left stick X/Y, right stick X/Y, L2 and R2. One press freezes, the next releases.

Axis windows

Click a stick's screen, the TRIGGERS screen or the TOUCHPAD screen to edit its axes together. Click a single pill (such as *X CC 40*) to edit that axis alone.



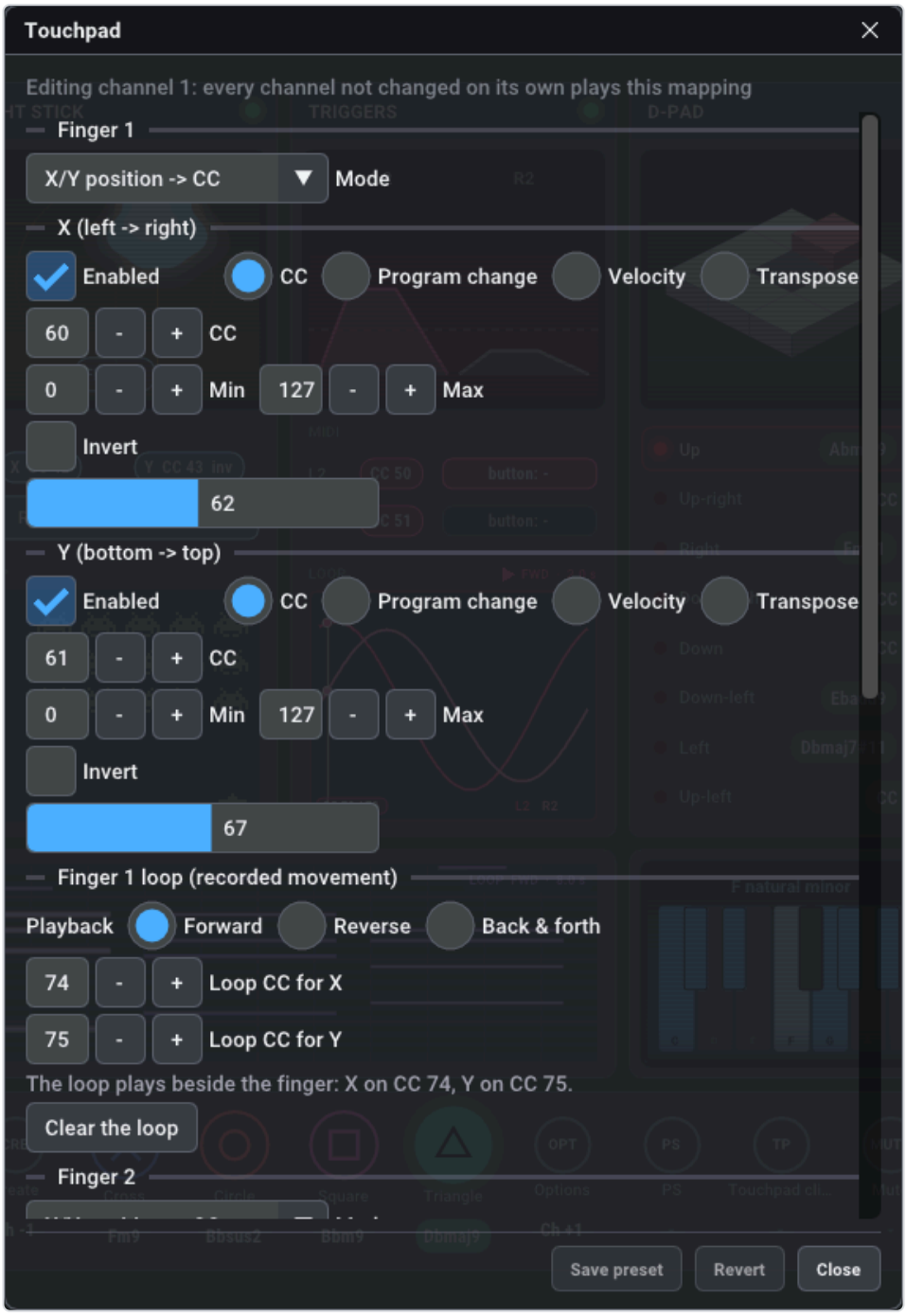
Each axis has these settings:

- **Enabled:** untick to silence the axis.
- **CC, Program change, Velocity or Transpose:** what the axis does. Program change is described in chapter 10, Velocity and Transpose below. The fields that follow are those of the CC mode.
- **CC:** the controller number (0–127).
- **Min and Max:** the value sent at each end of the travel. A Min above Max reverses the direction.
- **Invert:** flips the travel, for example so that pushing a stick up raises the value.
- **The bar** under the fields shows the value being sent right now.

A stick sends its minimum at one edge, its maximum at the other, and the middle of the range at rest.



The stick, trigger and touchpad windows also hold the settings of their loop (**Playback**, loop CCs, a lane per axis loop to change its length, **Steps**, **Clear the loop**), described in chapter 9.

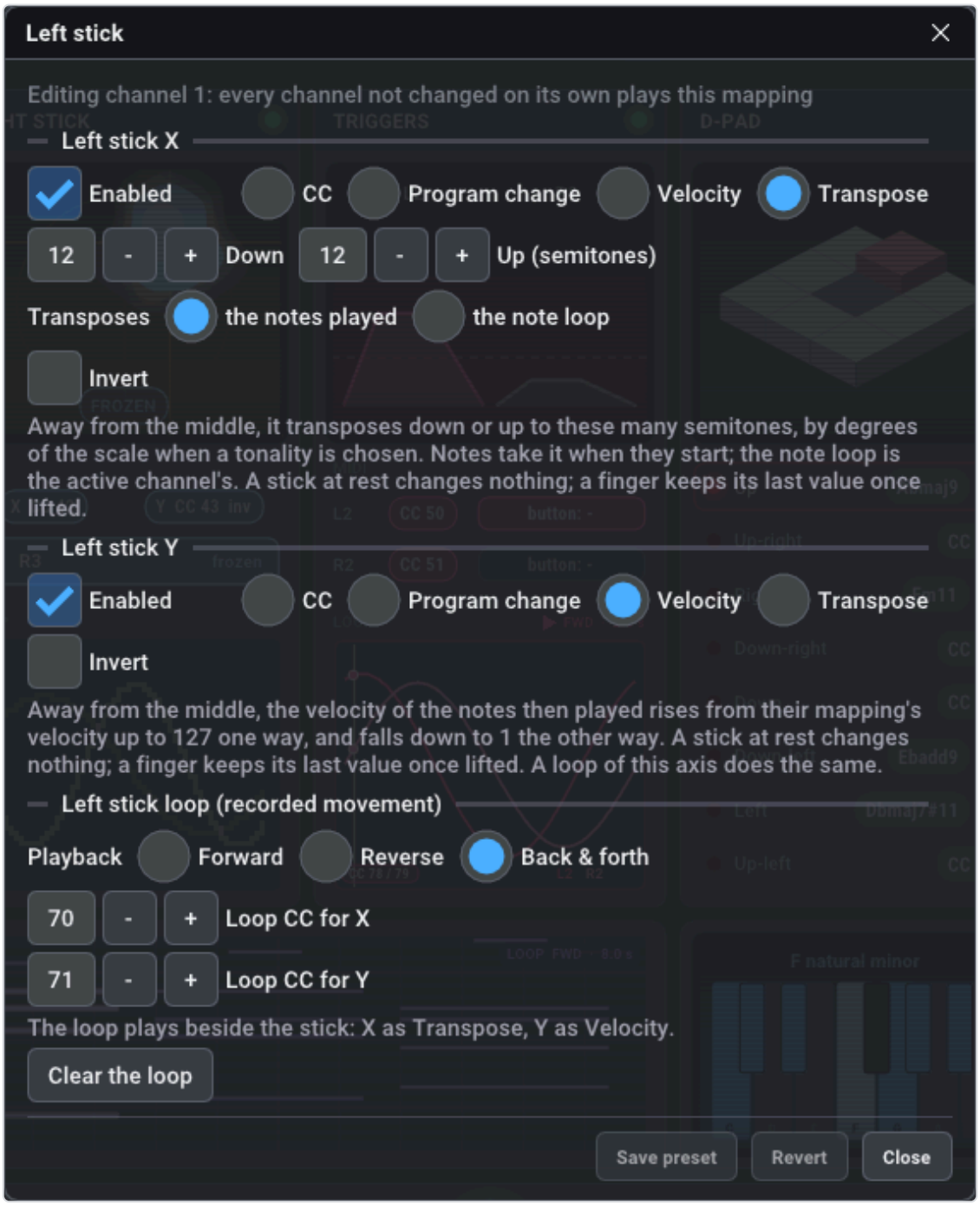


In the touchpad window, choose each finger's **Mode** first: *Off*, *X/Y position → CC* (two axes, as above) or *4 zones → pads* (a button action for each quarter: top left, top right, bottom left, bottom right). The window scrolls to show finger 2.

> **Note:** when a loop uses the same CC as a live axis, the window shows a red warning: both would send to the same parameter and fight. Choose different CCs.

Velocity and Transpose axes

An axis does not have to send a CC: set it to **Velocity** or **Transpose** and it shapes the notes you play instead. Choose the mode with the radio buttons at the top of the axis form.



Velocity

The axis changes the velocity of the notes then played, starting from the velocity set in their mapping:

- **Stick or finger:** away from the middle one way, the velocity rises towards 127 in proportion to the room left (halfway, a note mapped at 100 plays at 114); the other way, it falls towards 1.

- **Trigger:** pressing it only raises the velocity.
- It applies to the notes played live, not to the note loop.

Transpose

The axis transposes the notes then played, or the note loop:

- **Down** and **Up** set how far it goes at each end of its travel, from 0 to 24 semitones each way. A trigger has **Up** only.
- **Transposes the notes played / the note loop** chooses what it moves. The note loop is the active channel's.
- With a tonality chosen (chapter 7), the axis moves **by degrees of the scale**: 12 semitones make one octave of degrees, notes stay in the key and chords stay chords of the key (in F minor, one degree up turns Fm into Gdim). Without a tonality, it moves by semitones. A note outside the scale first goes to the nearest note of the scale.

What both have in common

- A note takes the value when it starts, and always stops at the pitch it started at.
- A stick at rest changes nothing: it has a small dead zone in the middle.
- A finger on the touchpad keeps its last value once lifted: set a velocity or a transposition, take your finger off, and it stays.
- A frozen axis (Axis freeze) keeps the value it had when it froze.
- A loop of such an axis plays the velocity or transposition instead of CCs: record a stick moving and the transposition follows the recorded gesture.
- Several axes set to the same mode add up.
- The axis pills show the live value: *VEL +40%*, *TR +3*, or *TR LOOP* for an axis set on the note loop.

> **Note:** in headless mode there is no tonality, so Transpose always moves by semitones.

7. Notes, chords and scales

MidiPad plays named chords such as *Fm9* or *Dbmaj7#11*, and can restrict every note and chord menu to the notes of a chosen scale, so you build a playable layout without leaving the key.

Notes

A **Note** action plays one MIDI note. Without a tonality, type its number (0–127); MidiPad shows its name beside it (60 = C4). With a tonality, the field becomes a menu of the notes of the scale on every octave, and the arrows step to the previous or next note of the scale.

Chords



A **Chord** action plays several notes at once, all with the same velocity:

- **Chord:** type a chord name in **Free entry**, or pick one in **Chord of the key** when a tonality is set.
- **Notes:** the notes the chord will play, spelled out, so you can check it before playing it.
- **Octave:** the octave of the root (4 by default: C4 = 60). The other notes stack above it.

- **Bass root:** *None*, *1 octave below* or *2 octaves below*. Doubles the root lower down to anchor the bass, as on the Cross button of the demo, where Fm9 is played over a low F.

A chord whose notes would fall outside the MIDI range (0–127) is refused; lower the octave.

Writing chord names

A chord name is a root (C, C#, Db ... B) followed by a type. MidiPad recognises these types:

Family	Types
Triads	(major), m, 5, dim, aug, sus2, sus4
Sixths	6, m6, 69, m69
Sevenths	7, maj7, m7, mmaj7, dim7, m7b5, 7sus4, 7#5, 7b5, maj7#5
Ninths	add9, 9, maj9, m9, mmaj9, 9sus4, m9b5, 7#9, 7b9
Elevenths	11, maj11, m11, m11b5, maj7#11, 9#11
Thirteenths	9b13, 13, maj13, m13

Useful details:

- A capital **M** means major: *CM7* is Cmaj7, not Cm7.
- *C6/9* is accepted for C69.
- A slash chord such as *Ab/C* plays the bass note one octave below the root's octave.

Tonality

Click the **SCALE** block to choose a tonality: a root on the left, a scale on the right. The tonality is shared by all channels and saved with your settings, not with the preset.



Available scales: major, natural minor, harmonic minor, melodic minor, dorian, phrygian, lydian, mixolydian, locrian, major pentatonic, minor pentatonic, blues and whole tone. Choose **None** as the root to remove the tonality.

With a tonality set:

- **Note** menus list only the notes of the scale.
- **Chord of the key** lists, degree by degree, every chord whose notes all belong to the scale.
- **Free entry** stays available for anything else. A note outside the scale is marked *out of key*; a chord with such notes shows *Contains notes outside* followed by the tonality. Both still play. An unknown chord name shows *Unknown chord*, and the previous chord stays active.
- The SCALE block shows the name of the tonality and lights its notes on the keyboard.

> **Tip:** set the tonality before building a preset. The Chord of the key menu then offers only chords that fit, which is the fastest way to fill the D-pad and the face buttons.

8. Channels

MidiPad sends on one MIDI channel at a time, the active channel, and you can switch channels while playing. Point each channel to a different instrument in your DAW and one controller plays a whole band: pads on channel 1, bass on channel 2, drums on channel 16.

Switching channels

- **In the window:** click the arrows of the channel selector in the top bar.
- **From the controller:** give a button the **Channel change** action, Previous (-1) or Next (+1). In the demo, Create steps down and Options steps up. Channels wrap around from 16 to 1.

The CHANNEL block shows the active channel in large digits. Its grid of sixteen channels marks with a violet dot every channel where a loop is playing.

Channel 1's mapping and channel mappings

Channel 1's mapping is the preset's main mapping: every channel plays it unless it has a **mapping of its own**. A channel's own mapping replaces channel 1's entirely while that channel is active: when you switch to it, every control takes its new meaning at once.

The demo preset uses this to give each channel its own role:

Channel	Mapping	Role in the demo
1	Base	Chords on the face buttons and D-pad, CCs on sticks and touchpad
2	Own	Other chords, notes on L2/R2
3–15	Base	Same as channel 1
16	Own	Single notes, touchpad split into four drum pads

On channel 1, the mapping windows edit channel 1's mapping, and every channel without its own follows. On channels 2 to 16, a channel plays channel 1's mapping until you change something there: that first change gives it its own mapping, a copy of channel 1's with your change, and from then on it no longer follows channel 1. The scope line at the top of each window tells you which one you are editing. Save and Save as keep every channel's own mapping.

> **Note:** to make a channel follow channel 1 again, open a mapping window on that channel, click **Use channel 1's mapping again** at the top and confirm: its own mapping is removed. Revert brings it back until you save the preset.

Shift and Loop record on every channel

A button, D-pad direction or touchpad pad set to **Shift** or **Loop record** on channel 1 keeps that role on every channel, even on a channel with its own mapping: you can always record, and always reach the Shift layer, whichever channel is active. On another channel, the mapping window of that control says *On every channel, this control is channel 1's Shift* (or *Loop record*) and stays greyed out: change it on channel 1. The layer itself still belongs to each channel: on channel 2, Shift brings up channel 2's Shift layer.

Notes never get stuck

A note always stops on the channel where it started. You can switch channels, or even presets, while holding a chord: its release still reaches the right instrument.

9. Loops and the piano roll

MidiPad records what you play and plays it back in a loop, on the channel where you played it, while your hands stay free for something else. It records two kinds of loops: **movement loops** (sticks, triggers, touchpad fingers), which work like hand-drawn LFOs, and **note loops**, which repeat the notes and chords you played.

Everything starts from a button set to **Loop record** (R1, labelled *Rec*, in the demo). Set on channel 1, it records on every channel (chapter 8).



Recording a movement loop

1. Hold **Loop record**.
2. Move a stick, press a trigger or slide a finger on the touchpad. You can move several at once.
3. Release **Loop record**. Each movement starts looping at once, from the start.

Only what moves is recorded: a stick must travel about 12 % from the centre to count. **Each axis records on its own:** a stick's left-right (X) and up-down (Y), a finger's sideways and up-down moves, L2 and R2 are separate loops. Each starts when its axis moves, so the two can have different lengths, and you can record one axis now and the other one later. An axis that is not **Enabled** in its window is never recorded. For the touchpad, the moments when the finger was lifted are recorded too, and play nothing. A loop lasts up to 60 seconds.

The loop plays **beside** the control, on CCs of its own, so the stick or finger stays free to play live on its usual CCs. The loop CCs are set in the window of the stick, triggers or touchpad:

Source	Loop CCs (default)
Left stick	70 / 71

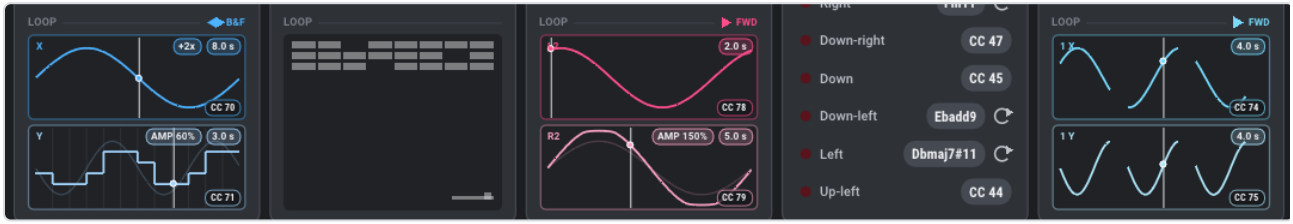
Source	Loop CCs (default)
Right stick	72 / 73
Touchpad finger 1	74 / 75
Touchpad finger 2	76 / 77
Triggers L2 / R2	78 / 79



In the same window, **Playback** chooses the direction: *Forward*, *Reverse* or *Back & forth*. **Clear X** and **Clear Y** (**Clear L2** and **Clear R2** for the triggers) erase one axis' loop, **Clear both** the two.

Reading the LOOP panels

The LOOP panel of each module shows the loops of the active channel, **one lane per axis**: X and Y for a stick, L2 and R2 for the triggers, 1 X and 1 Y for the touchpad (2 X and 2 Y too once finger 2 holds a loop). Each lane spans the whole width whatever the length of its loop, with its own playhead: a shorter loop is drawn just as wide, and its playhead crosses it faster. A lane shows its name at the top left, its length at the top right (*4.0 s*, or *1.5 BARS* for a loop synced to a clock; in red while it records) and its loop CC at the bottom right; the header shows the playback mode (FWD, REV or B&F). The curves show what the loop sends: the higher the value, the higher the curve, so an inverted axis is drawn upside down. The lane of an axis that is not enabled reads *not enabled · not recorded*. Click the panel to open its settings; right-click a lane for its menu: lock, clear, back to 1x · 100% (chapter 4).



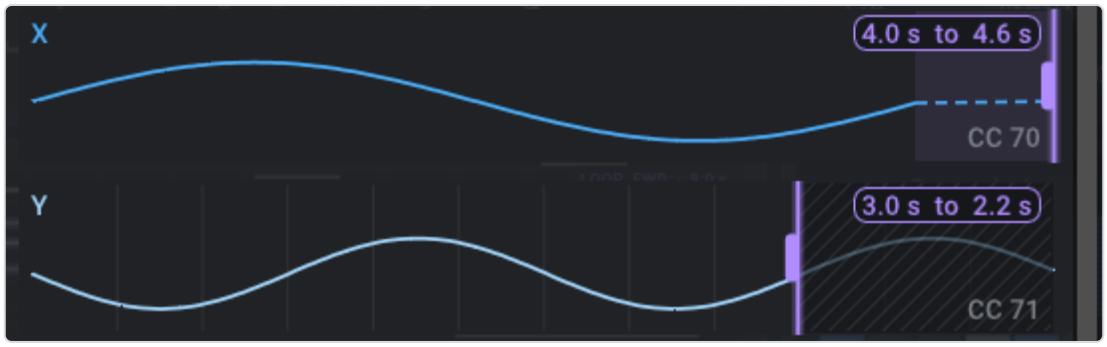
Over a lane, a small red **×** appears beside its name: click it to clear that axis' loop alone, the other one playing on. A locked lane has none. In the control's window, each lane has its × too.

An empty LOOP panel plays a small grey pixel animation, in the style of 1980s video games, so you can tell at a glance which loops are free. It is decoration only.

Axis loop lengths and steps

The window of a stick, the triggers or the touchpad (click its screen or its LOOP panel) shows, under its loop settings, a lane for each axis loop that plays, over its own length, and below them the step settings of each axis. Scroll the window down to reach them.

Shortening or lengthening an axis loop. Drag the violet handle at the right end of a lane, as in the note loop editor. Shorter, the part cut is hatched; longer, the time added is drawn as a dashed line from the loop's last value back to its first, so that the loop closes without a jump. The chip reads the old and the new length (*4.0 s to 4.6 s*); release to apply it, or press Escape to cancel. A loop synced to a clock keeps whole beats; a loop in steps, whole steps.



Loops in steps. Each axis has a **Steps** box. Ticked, its loop no longer glides: it holds one value per step and jumps to the next, like a step sequencer. **Levels** (2 to 128) sets how many values it may take, each rounded to the nearest; **Step** sets how long each value lasts, as a note value: 1/32, 1/16, 1/8, 1/4 (a beat), 1/2 or 1 bar. The beat is that of the loop when it is synced to a clock, otherwise the tempo shown in the top bar.

Steps apply as the loop plays: what was recorded stays as it was, so you can change them or turn them off at any time. The lane draws the staircase the loop plays, the recorded curve faintly behind it, and the step grid. Steps are saved in the preset, with each mapping's loops.

Recording a note loop

1. Hold **Loop record**.
2. Play notes or chords: face buttons, D-pad, triggers used as buttons, touchpad pads.
3. Release **Loop record**. The notes loop on the channel where you played them, with their velocity and humanization.

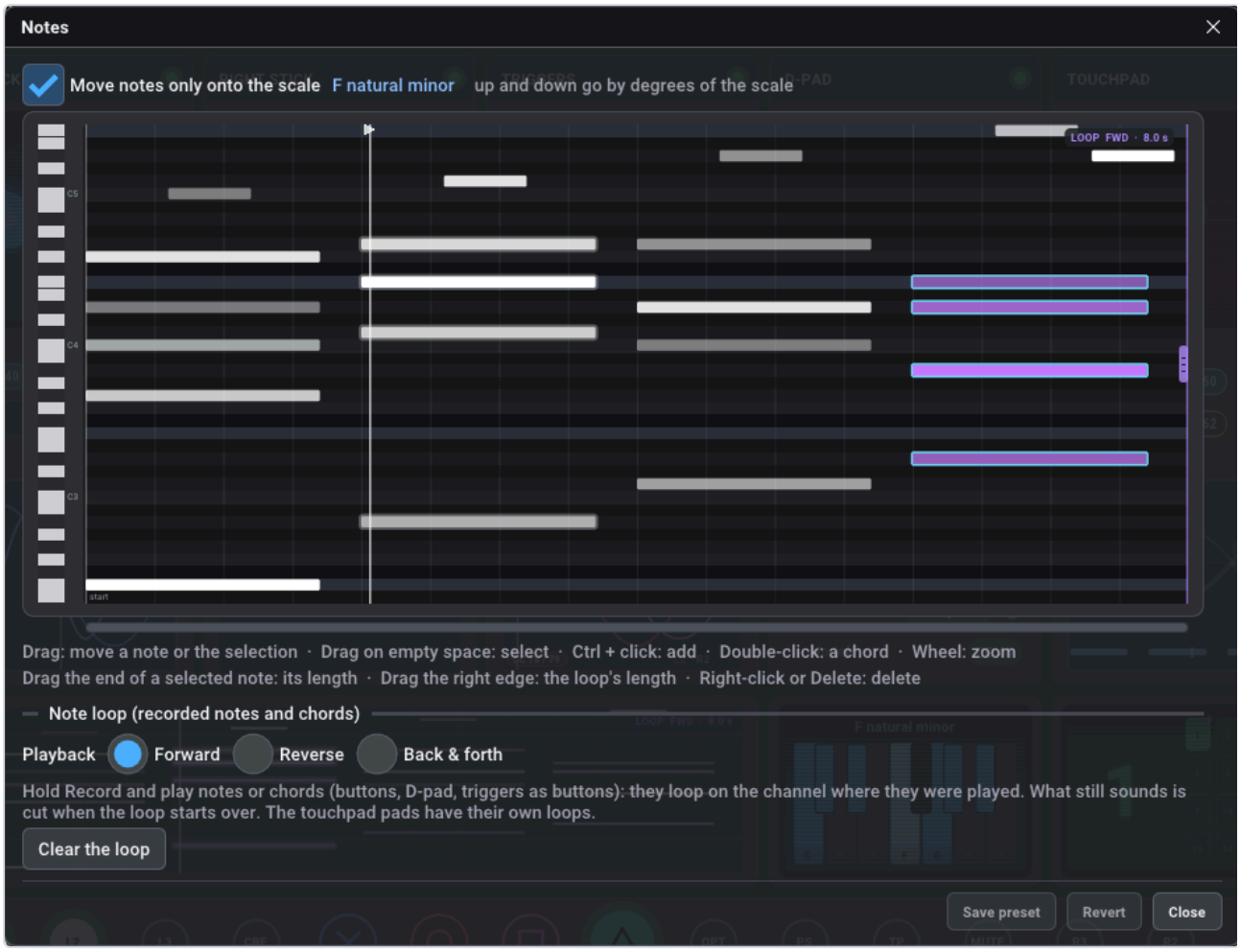
Each channel has one note loop. Recording a new one replaces it. A note still held at the end of the recording (a latched chord) sounds until the end of the loop, and everything sounding stops when the loop starts over.

The piano roll

Without a note loop, the piano roll scrolls the last seconds of the active channel towards the *now* line on the right edge: white for what you play, violet for what a loop plays. Its keyboard follows the range of the notes played.

While a note loop plays, the piano roll becomes a **clip**: the whole loop, from *start* to its end, standing still, with a playhead sweeping it in the playback direction. Its notes are white, as opaque as they are loud, and get a violet halo as the playhead crosses them. Beneath the clip, the notes you play live keep scrolling from the right edge in faint cyan, as without a loop. Click the piano roll to open the note loop editor: a large piano roll to edit the notes (below), the playback mode and Clear the loop.





Editing the notes of a loop

The note loop editor, opened by a click on the piano roll, holds a large piano roll of the loop. Edit its notes there with the mouse, without recording again, while the loop keeps playing; the piano roll of the main view only shows the loop.

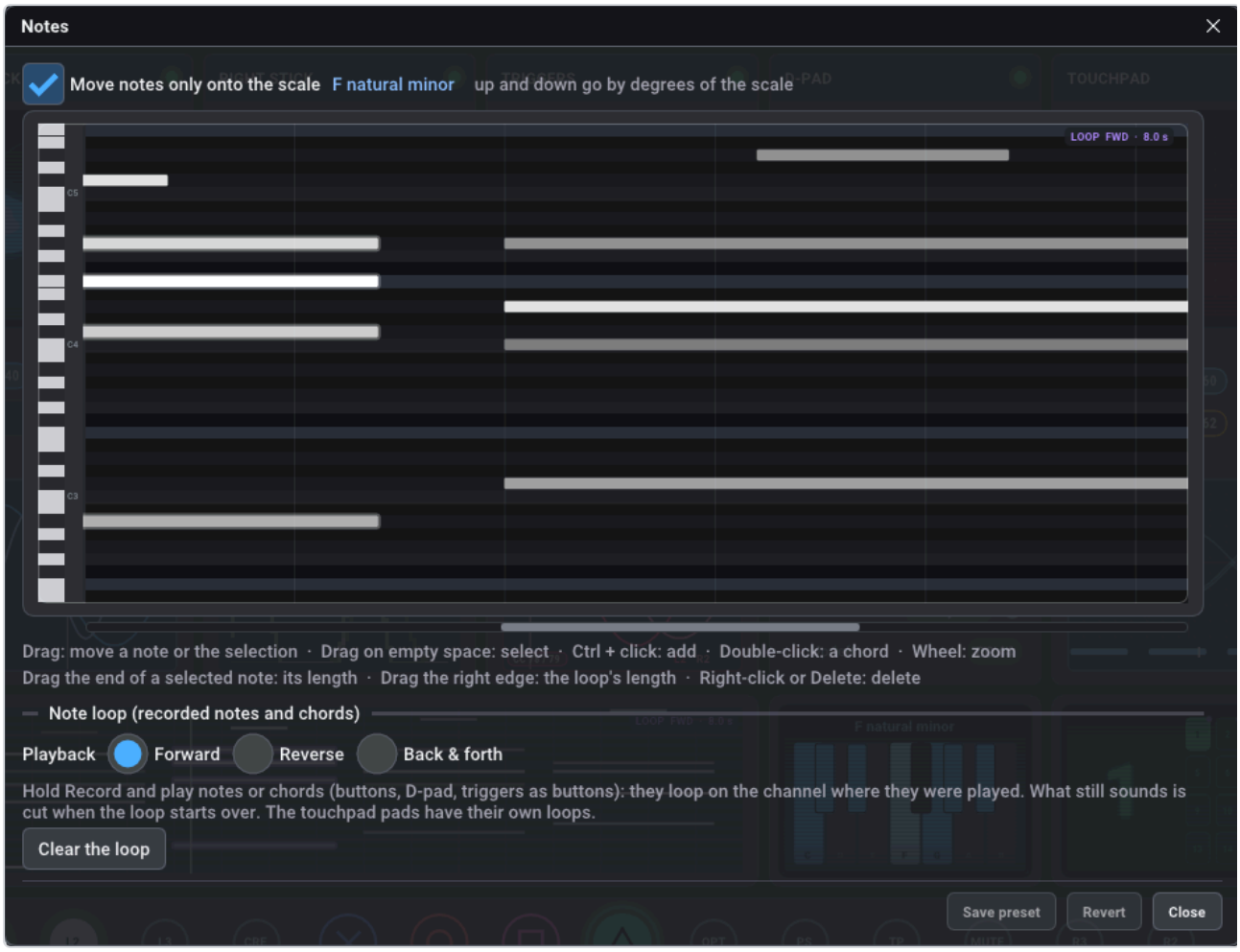
Gesture	What it does
Drag a note	Moves it, in time and pitch, its length kept; on a selected note, moves the whole selection as a block
Drag over empty space	Selects the notes in a rectangle (with Ctrl, adds them to the selection)

Gesture	What it does
Ctrl + click a note	Adds it to the selection, or takes it out
Double-click a note	Selects its chord: every note of the same press, even spread by humanization
Drag the end of a selected note	Lengthens or shortens every selected note alike
Right-click a note, or Delete	Deletes it, or the whole selection it belongs to
Click on empty space, or Escape	Clears the selection

Notes are white, as opaque as they are loud: solid white at velocity 127, faint near 0. Selected notes are mauve, shaded the same way and outlined in cyan; notes the playhead crosses have a white glow.

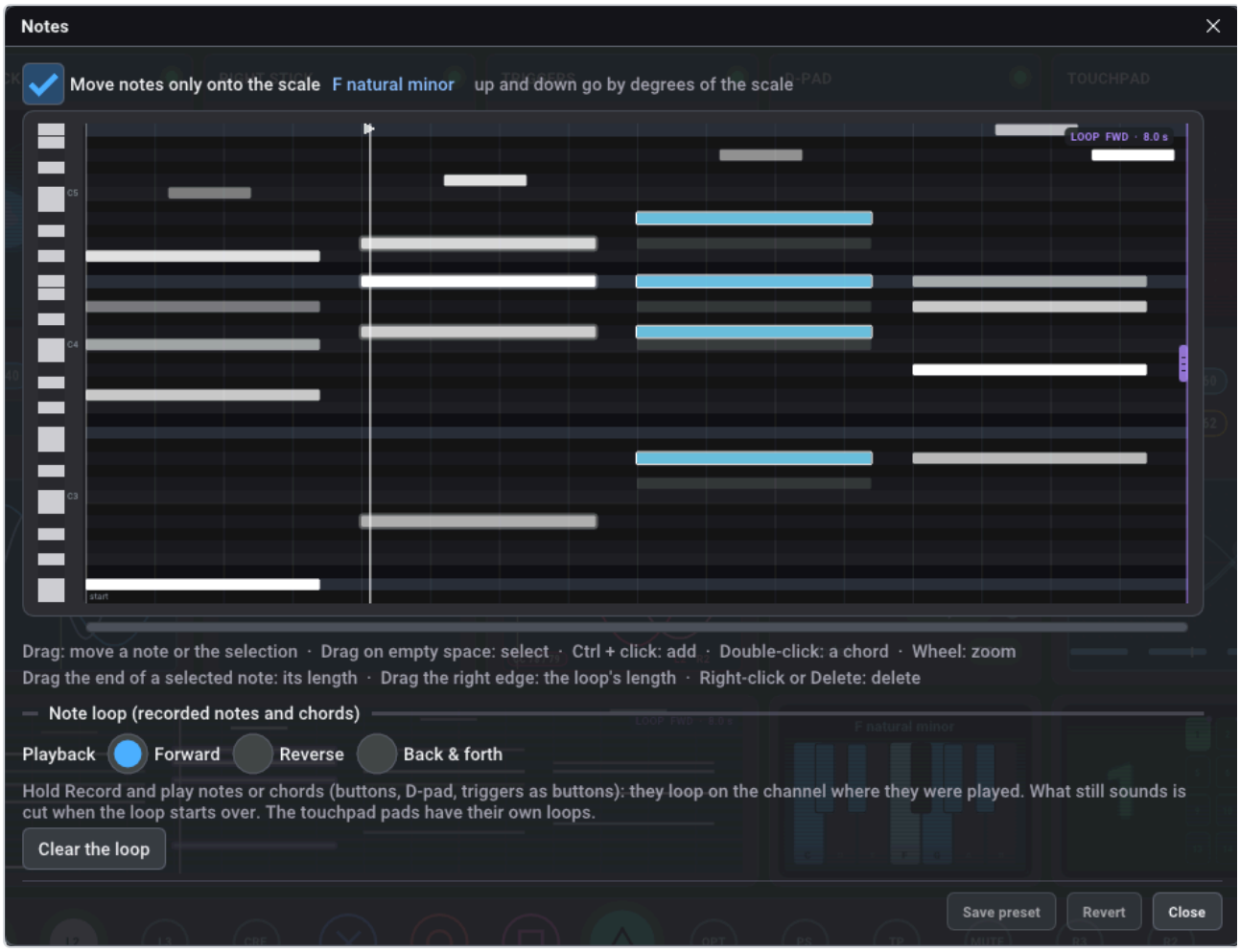
Zoom

Hold **Ctrl** and turn the mouse wheel over the editor's piano roll to zoom in or out around the mouse, from the whole loop to a second (or a beat) across; the wheel alone, or a sideways scroll, moves along the loop. The scrollbar under the roll shows the part in view: drag its thumb, or click beside it to go there. Every gesture works zoomed in, and the loop's end handle shows when the end of the loop is in view.



Moving on the scale

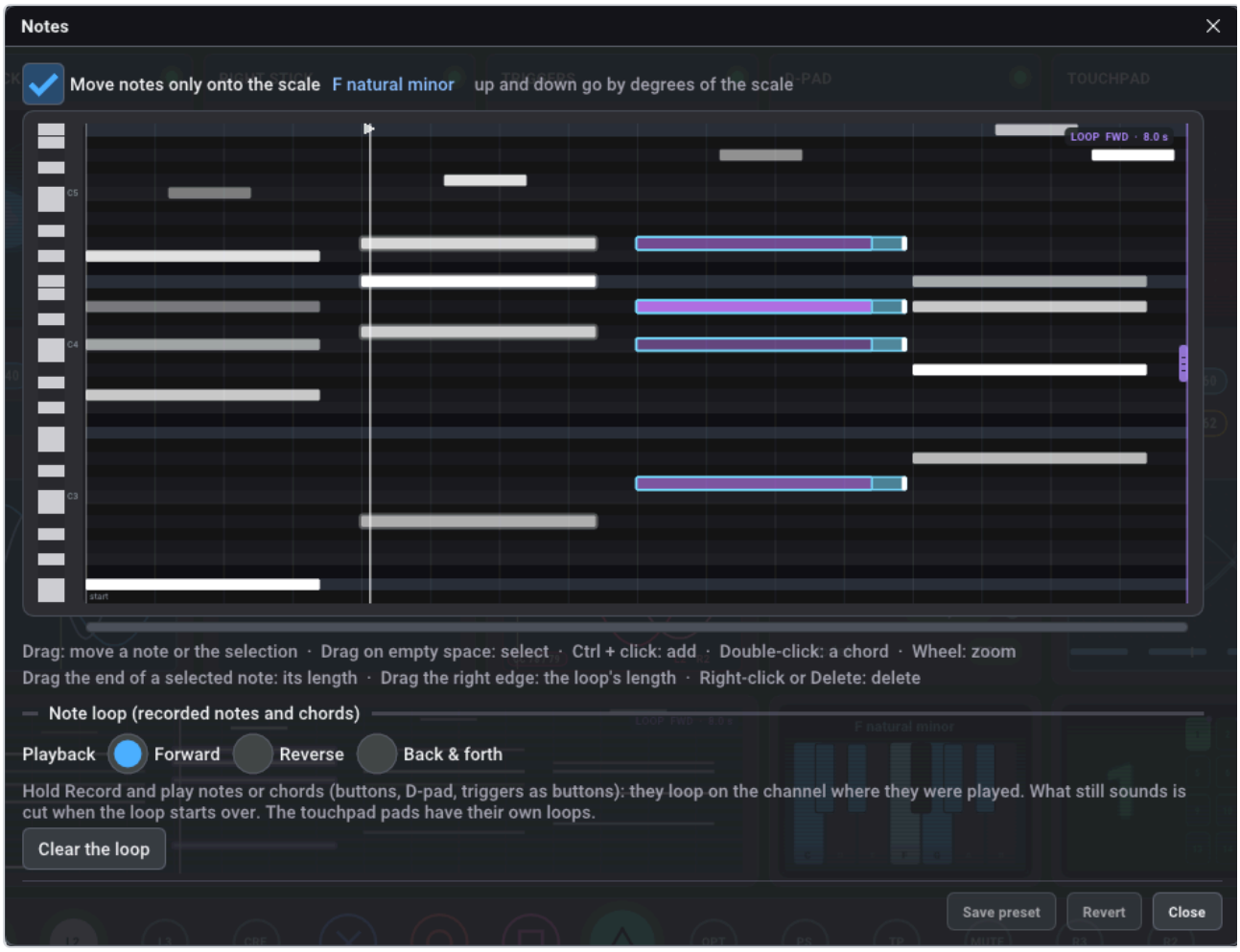
With a tonality chosen, **Move notes only onto the scale** (ticked by default, at the top of the editor) keeps moved notes in key: up and down go by degrees of the scale, so each note of a chord moves by its own interval, and a note outside the scale goes to the nearest note of it. Rows outside the scale are darkened, and the root's rows tinted blue. Untick it to move by semitones.



While you drag, cyan copies show where the notes go; release the mouse button to move them. A group keeps the exact gaps between its notes, so a humanized chord keeps its feel. With a clock and the quantizer on, the group's first note snaps to the grid and the others follow. The group stays inside the loop as a block; notes of the same pitch it lands on are cut short, and notes sounding stop cleanly.

Note lengths

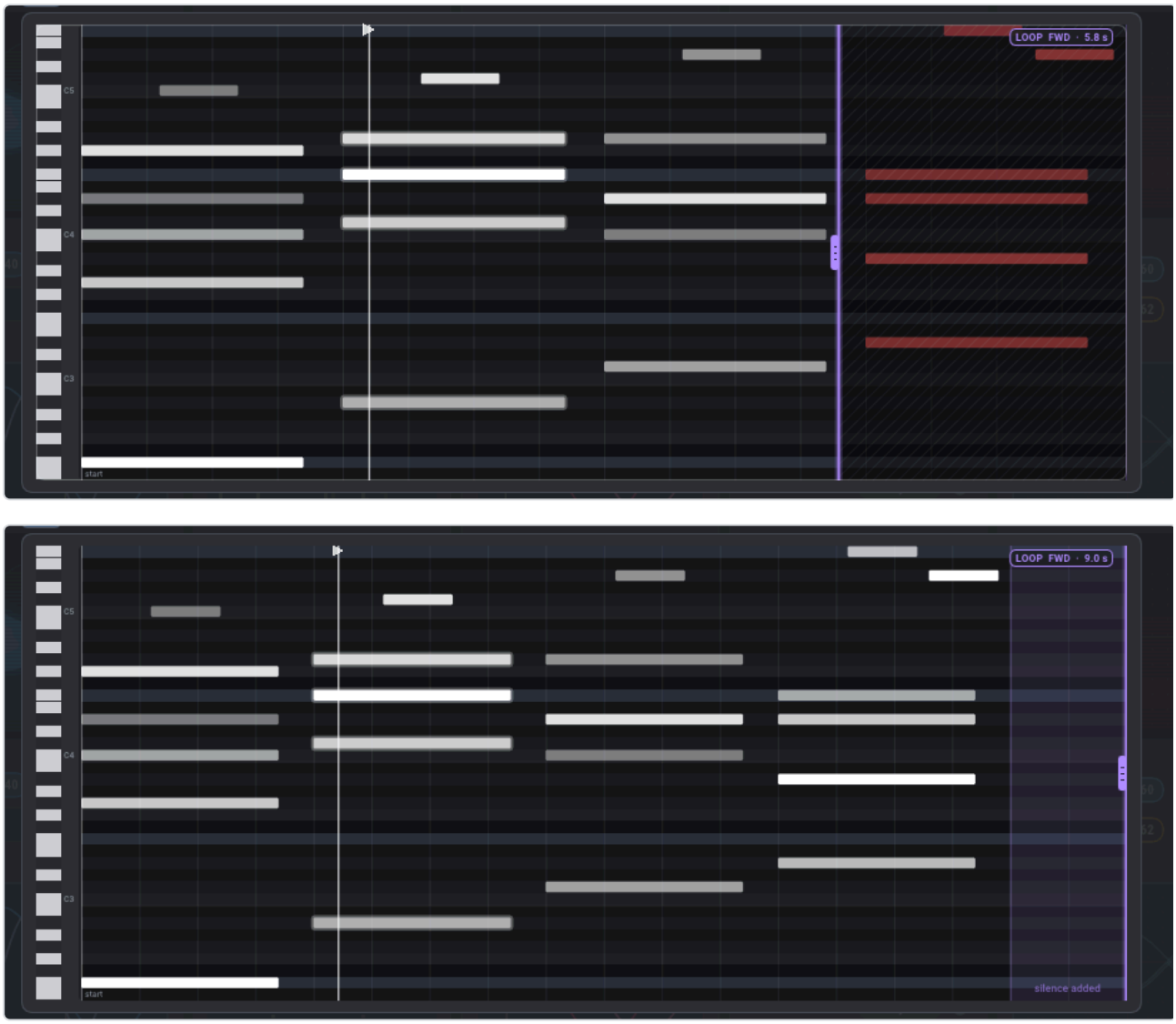
Point at the right end of a selected note: the cursor becomes a double arrow. Drag it to lengthen or shorten every selected note by the same amount; while you drag, cyan shows where the notes will end. With the quantizer, the grabbed end snaps to the grid. Each note keeps a minimal length, ends inside the loop, and stops where the next note of its pitch starts.



Loop length

The violet bar at the right edge of the editor's piano roll is the end of the loop; its grip sits in the middle. Point at it: the cursor becomes a double arrow. Drag it to correct a loop whose Loop record was released too late, or to make room for overdubs.

- **Drag left** to shorten the loop. The part that goes is dimmed and hatched, and the notes that start there turn red: they will be deleted. Notes still on at the new end end there.
- **Drag right** to lengthen it, up to a minute. The bar stays at the edge and the loop narrows to make room; the silence added is tinted violet. The mouse may leave the piano roll, and the window, while you drag.



The chip at the top right shows the new length while you drag. Release the mouse button to apply it, or press Escape to keep the loop as it was. A loop synced to a clock changes by whole beats, one at least, so it stays in time with the other loops; a free loop changes freely. What sounds stops cleanly, and the playhead starts over if it was past the new end. There is no undo: a shortened loop's notes are gone.

Overdub

Overdub adds notes to a note loop without changing its length.

1. While a note loop plays, tap **Loop record** briefly (less than half a second) without playing anything: overdub is on.
2. Play: every note is added to the loop where it plays.
3. Tap **Loop record** again to turn overdub off.

During overdub, the piano roll shows **OVERDUB** and an orange playhead, each added note appears in orange while you hold it, and the Rec pad is circled in orange.

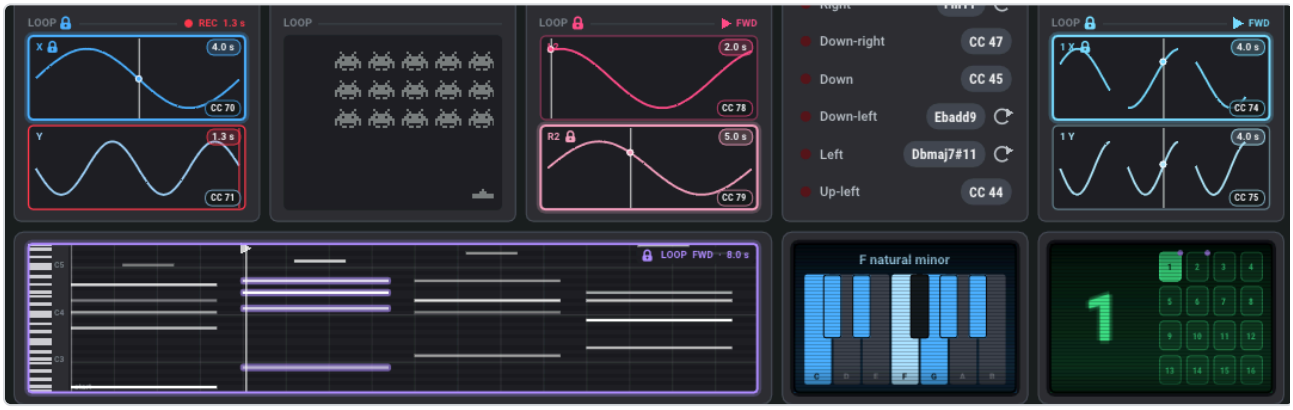


Stopping and restarting loops

You do	Result
Tap Loop record, no note loop playing, nothing moved	Stops the loops of the active channel, except the locked ones
Hold Loop record without playing or moving	Stops the loops of the active channel, except the locked ones
Click ■ (Stop) in the top bar	Pauses the loops of every channel, without erasing them
Click ▶ (Play)	Restarts every loop from its start; Play turns green while loops run
Record while loops are paused	Restarts all loops together with the new one
Click PANIC	Stops every loop on every channel, the locked ones too, and silences everything

Locking a loop

A locked loop stays: pressing Loop record again neither replaces it nor stops it, so you can go on recording the other controls, or tap and hold Loop record freely, without losing it. **Each axis locks on its own:** lock a stick's X loop and record its Y later; the locked one stays as it is.



- **With the mouse:** right-click a lane of a LOOP panel and choose **Lock** (or **Unlock**) for that axis' loop; right-click the panel's header for all of its axes (an axis without a loop stays free to record); right-click the piano roll for the note loop. Only a recorded loop locks.
- **From the controller, with the menus:** in the edit mode (Shift + Mode), put the cursor on a lane or the piano roll and press Menu (chapter 4).
- **From the controller, while playing, a movement loop:** with Shift on (held, or latched in Toggle mode), hold Loop record and move the axis whose loop you want to lock or unlock: a stick left-right for its X, up-down for its Y, L2 or R2, or a finger on the touchpad (axis mode) sliding sideways or up and down once it has landed. Each axis changes once per press of Loop record. Meanwhile the controls neither stretch loops nor send CCs, and releasing Loop record does nothing else.
- **From the controller, while playing, the note loop:** with Shift on, tap Loop record twice quickly (within 0.35 s), playing nothing in between. The overdub the first tap turned on is taken back; when the first tap would stop the loops, it waits 0.35 s for a second one.

A locked lane shows a padlock beside its name and a solid outline in its colour, and the panel a padlock next to LOOP (on the piano roll, beside its LOOP chip). The lock belongs to the loop of the active channel; in 4-pad mode the touchpad panel shows the note loop and locks it.

A locked note loop takes no overdub. ■ and ▶ pause and restart locked loops as usual, Shift still stretches and widens them, and you can still edit the notes or use **Clear the loop**. PANIC clears every loop, the locked ones too. Sessions keep which loops are locked.

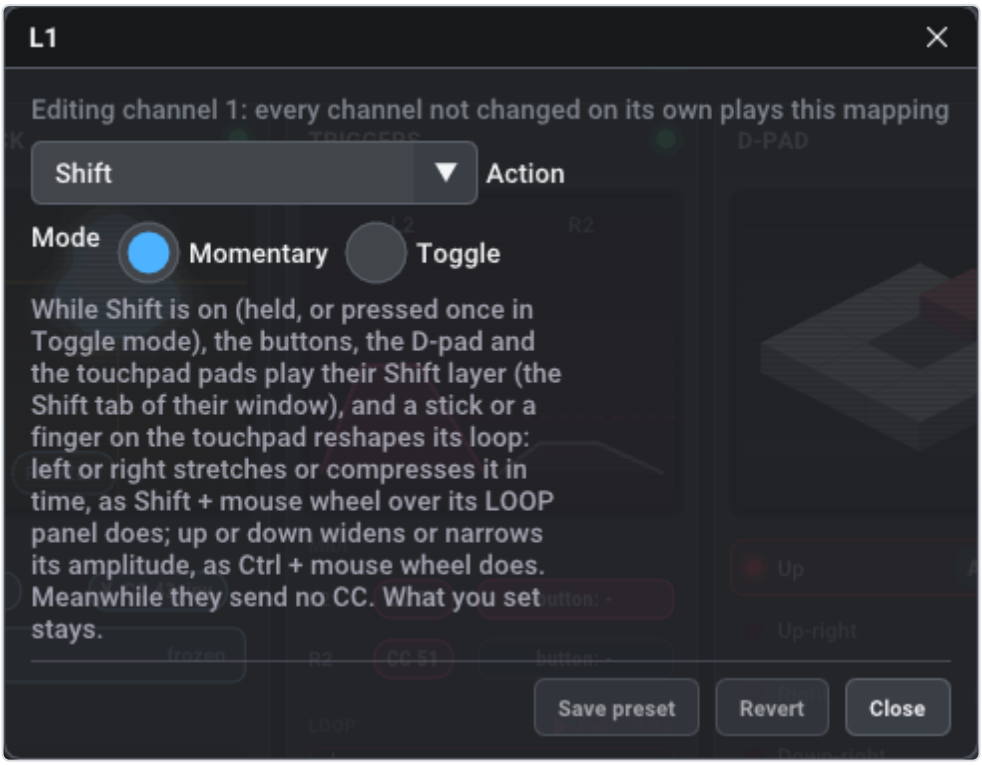
Stretching loops in time and amplitude

A movement loop (sticks, triggers, touchpad in axis mode) can be reshaped while it plays, without recording it again, right from the gamepad: stretched or compressed **in time**, and narrowed **in amplitude**. The note loop is not concerned.

Shift: hold a button, move an axis

Give a button or a D-pad direction the **Shift** action. Hold it (or, in Toggle mode, press it once), then move the stick or the finger whose loops you want to reshape. Each axis loop changes on its own:

- **Stick left or right** compresses or stretches its **X** loop **in time**, **up or down** its **Y** loop: up to **100 times longer** (*+100x*), up to **20 times shorter** (*-20x*), by steps of a half (*+1.5x*, *+2x*, *+2.5x* ...). Right and up lengthen.
- **Click the stick (L3 or R3) while Shift is on** to switch it to the **amplitudes**: left or right then widens or narrows X's, up or down Y's, around the loop's middle (halfway between its lowest and highest values): from **0 %** (still) through **100 %** (as recorded) up to **800 %**, by steps of 5 % below 100 % and of 25 % above. Click again to go back to the lengths. While Shift is on, a **LEN** or **AMP** badge at the top right of the stick's screen tells which it changes.
- Widened past 100 %, a swing would run beyond the ends of the CC range (0 and 127), so MidiPad **clips it softly**: the curve bends towards the end, then stays flat there, like an LFO driven into saturation. The bend grows with the amplitude, from none at 100 % to a tenth of the range from 200 %. A small swing widened 3 or 4 times fills the whole range; a wide one widened far turns into a square wave.
- The stick works as a wheel kept turning: nothing in the dead zone around the centre, then faster the further you push. Only the way it mostly moves counts, so a diagonal never changes both at once.
- **On the touchpad** (axis mode), slide a finger: each short slide counts one step, sideways for the X loops, up and down for the Y loops. Click the touchpad while Shift is on to switch between lengths and amplitudes (LEN or AMP on its screen).
- **The triggers'** loops (L2, R2) are reshaped with the mouse (below).
- Crossing from compressed to stretched, the length waits a moment at 1x (and the amplitude at 100 %, crossing it either way), so the recorded loop is easy to find again.
- Release the button (in Toggle mode, press it again): everything you set stays. Recording the loop again puts it back to 1x and 100 %.

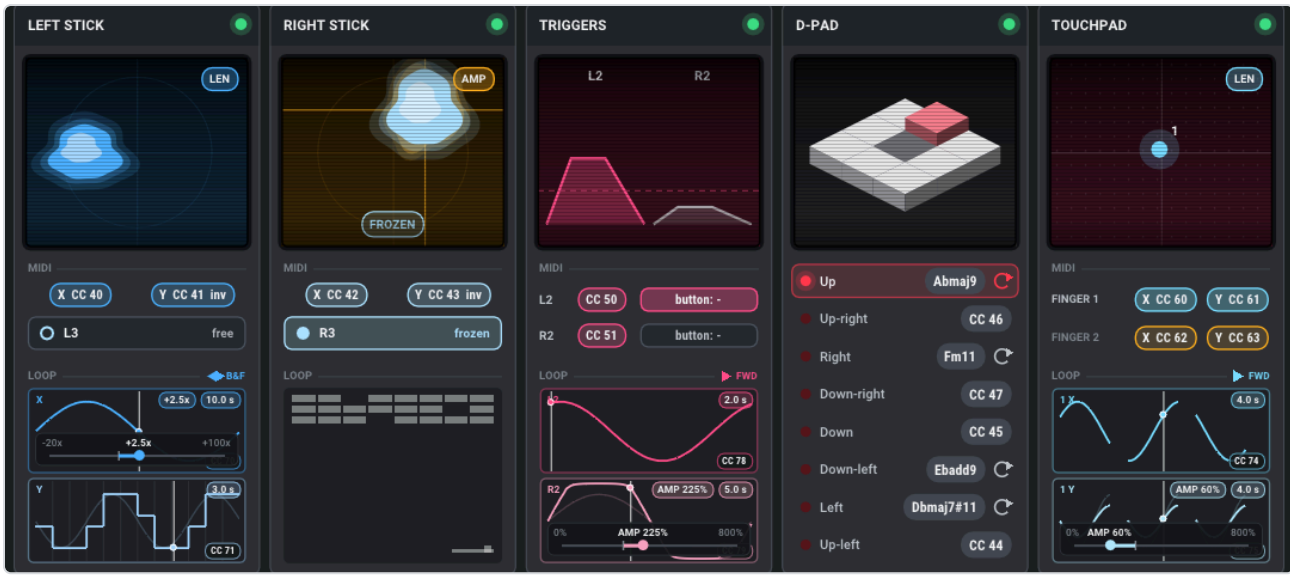


While Shift is on, the sticks and fingers only reshape loops: they send no CC and start no recording; the gauges of the lanes show what changes (below).

The loop goes on from the same point when it changes, so it never jumps. Stretched, it is smoothed between its recorded points; compressed, each value is the average of the points it passes over; a change of amplitude is eased in over a few milliseconds. A loop of a Velocity or Transpose axis is reshaped the same way.

What you see

While the length or the amplitude of a loop changes, a gauge appears at the bottom of its lane: a log scale from -20x to +100x for the length, a scale from 0 % to 800 % for the amplitude, with 100 % marked. It fades out a moment after the last change.



Once set, chips stay on the lane next to its length (*+2.5x*, *AMP 60%*), and the curve is drawn as it plays, narrowed or widened (flat where it is clipped), the recorded one faintly behind. The length shown is the stretched one (*10.0 s*, or *12:30 min* beyond a minute; in beats or bars for a loop synced to a clock).

At the computer

The same changes are also at hand with the mouse, over a lane of a LOOP panel: the **Shift** key + wheel for its length, **Ctrl** + wheel for its amplitude (up stretches or widens); over the panel's header, every lane at once. Turned quickly, the wheel takes bigger steps. This is also how the triggers' loops are reshaped.

Playing in time: internal clock or Ableton Link

By default MidiPad keeps no beat: a loop lasts exactly as long as you held the button. Give it a clock and loops come out a whole number of beats long, start on the beat and follow the tempo. There are two clocks: the **internal clock**, MidiPad's own tempo, with no network needed, and **Ableton Link**, which joins the tempo and beat of your DAW, or of any app using Link on the same network.

Choose the clock in ☰ › **Options** › SYNC › **Clock**. The clock button of the top bar then reads *CLOCK 92.5 BPM* or *LINK 120.0 BPM · 2* (with Link, the number of peers: the other Link apps in the session), with a square for each beat of the bar, the current one lit. Without a clock it reads *FREE*. A click on the button, away from the tempo, turns the clock off, and another turns the same clock back on. Alone on the network, a Link session starts at 120 BPM.



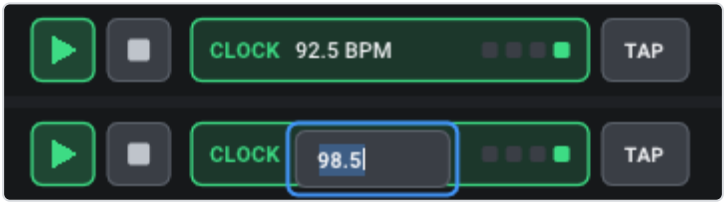
A loop recorded while a clock runs:

- **Snaps to whole beats.** When you release Loop record, the length is rounded to the nearest whole number of beats (or bars, see SYNC in Options), one at least. Hold for about six beats and you get exactly six: *1.5 BARS*. One beat, three beats, one bar, two and a half bars: anything goes.
- **Starts on the beat.** Its start moves to the nearest beat, so it stays in phase with the clock and with the other loops.
- **Follows the tempo.** It plays in beats, not in seconds: change the tempo, in MidiPad or with Link in your DAW, and the loop speeds up or slows down with it, still in phase.
- **Shows beats and bars.** Its header gives the length in beats or bars (*1 BEAT*, *3 BEATS*, *1 BAR*, *1.5 BARS*). The clip of the piano roll draws a grid of the beats, stronger on the bars, with the bar numbers on top; where the beats would crowd, only the bars, or every few bars.
- **Still stretches.** Shift works as before: *+2x* makes a one-bar loop last two bars, *-2x* half a bar. Forward, Reverse and B&F all work; in B&F each pass lasts the loop's length.

Note loops and movement loops behave alike.

Setting the tempo

The tempo belongs to the clock. The internal clock's is MidiPad's own and is kept for the next launch. With Link, any app in the session can change it and every other one follows: change it in MidiPad and your DAW follows, as MidiPad follows your DAW.



- **Mouse wheel over the tempo**: 1 BPM a notch, 0.1 with the Shift key; turned quickly, it takes bigger steps.
- **Double-click on the tempo**: type a value (for example 92.5), then Enter. Escape cancels.
- **TAP**, next to the clock button: tap in time; from the second tap, the tempo follows your taps. It takes the median of the last four intervals, so one tap clearly off is ignored; a pause of more than two seconds starts over.
- **Tap tempo on the controller**: give a button the *Tap tempo* action (chapter 6) to tap with the gamepad in your hands.

Tempos go from 20 to 999 BPM.

Quantizing notes

With a clock running, ☰ › Options › SYNC › **Quantize notes** (*Off*, *1/8*, *1/16*, *1/32*) puts the notes recorded into a note loop on a grid of eighth, sixteenth or thirty-second notes. What you hear while playing is never delayed: only the loop's copy moves.

- Each press moves as a whole: its first note goes to the nearest step, earlier or later, and its release moves with it, so the note keeps its length. A note played just before the loop's end goes to its start.
- A chord lands on the grid together. Humanized, its first note goes on the grid and its other notes keep their spread, so the chord stays alive.
- Notes added in overdub are quantized as you play them.
- Notes of the same pitch never overlap: one still sounding when the next starts ends there, and two landing on the same step become one.

Free loops and movement loops are never quantized.

Start and stop. With Link and *Start / stop sync* on (the default), ▶ and ■ also start and stop the Link session, and MidiPad's loops pause and resume when another app starts or stops it. With the internal clock, ▶ and ■ act on MidiPad's loops.

MIDI clock out. With the internal clock, MidiPad also sends MIDI Clock to its output: 24 pulses a beat at the clock's tempo, Start when ▶ starts the loops, and Stop when ■ stops them or when you leave the internal clock. Drum machines, sequencers, arpeggiators and DAWs set to follow an external MIDI

clock then play in time with MidiPad. It is on by default; turn it off in ☰ › Options › SYNC › MIDI clock out.

Free mode. Without a clock, loops keep the length you played, to the millisecond. A loop recorded free stays free when you turn a clock on; a synced loop goes on at its own pace when you turn the clock off.

> **Tip:** Link works over the local network. If MidiPad sees 0 peers while another app has Link on, check that both are on the same network and that no firewall blocks UDP multicast (Link uses port 20808).

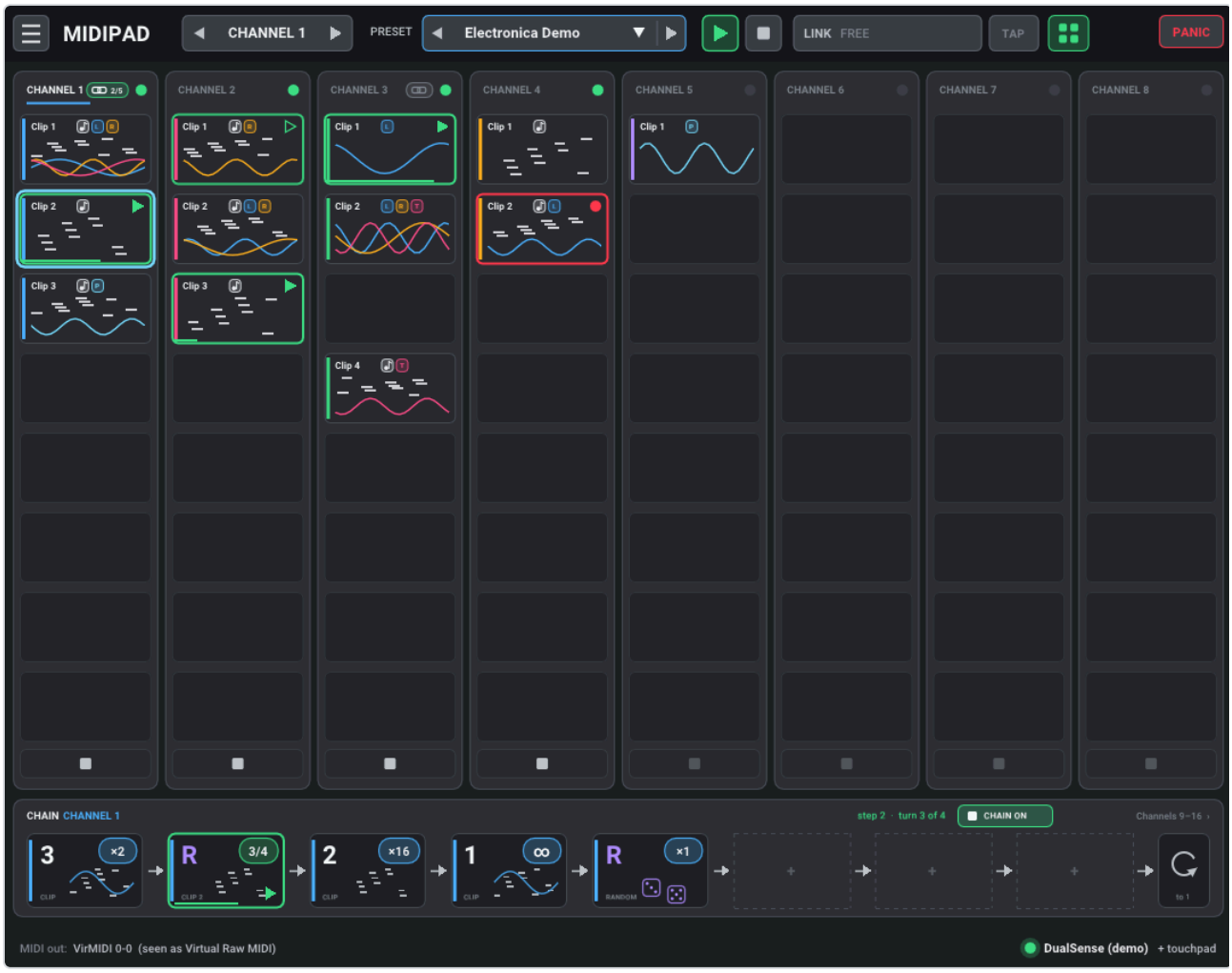
Loops on several channels

Each channel has its own loops. Record on channel 1, switch to channel 2, record again: all loops play together, each on its own channel. The LOOP panels and the piano roll always show the loops of the active channel; the CHANNEL block marks the others with a violet dot.

> **Note:** loops are not saved with the preset. Save them in a session (below) to find them again after quitting MidiPad.

Clips and the session screen

Each channel keeps up to eight **clips**. A clip holds all of the channel's loops at one moment: the note loop, both sticks, the triggers and the touchpad fingers, with their stretch, amplitude, sync and lock. A channel plays one clip at a time, independently of the others: clip 1 on channel 1 can play beside clip 3 on channel 2. Everything described so far in this chapter (recording, overdub, locking, the note loop editor, the LOOP panels) acts on the **current clip** of the active channel.



Opening the session screen

- **From the controller:** a press on the button given the **Mode** action (chapter 6). Set on channel 1, it works on every channel; without one in the preset, the PS / Guide button does it.
- **With the mouse:** the button with four squares in the top bar, right of TAP. **On the keyboard:** Tab.

The session screen takes the place of the modules: a column per channel, eight at a time (**Channels 9–16 ›** shows the others, and the screen follows the active channel), a row per clip slot, at the bottom of each column a square that stops the channel, and under the columns the chain of the active channel (below).

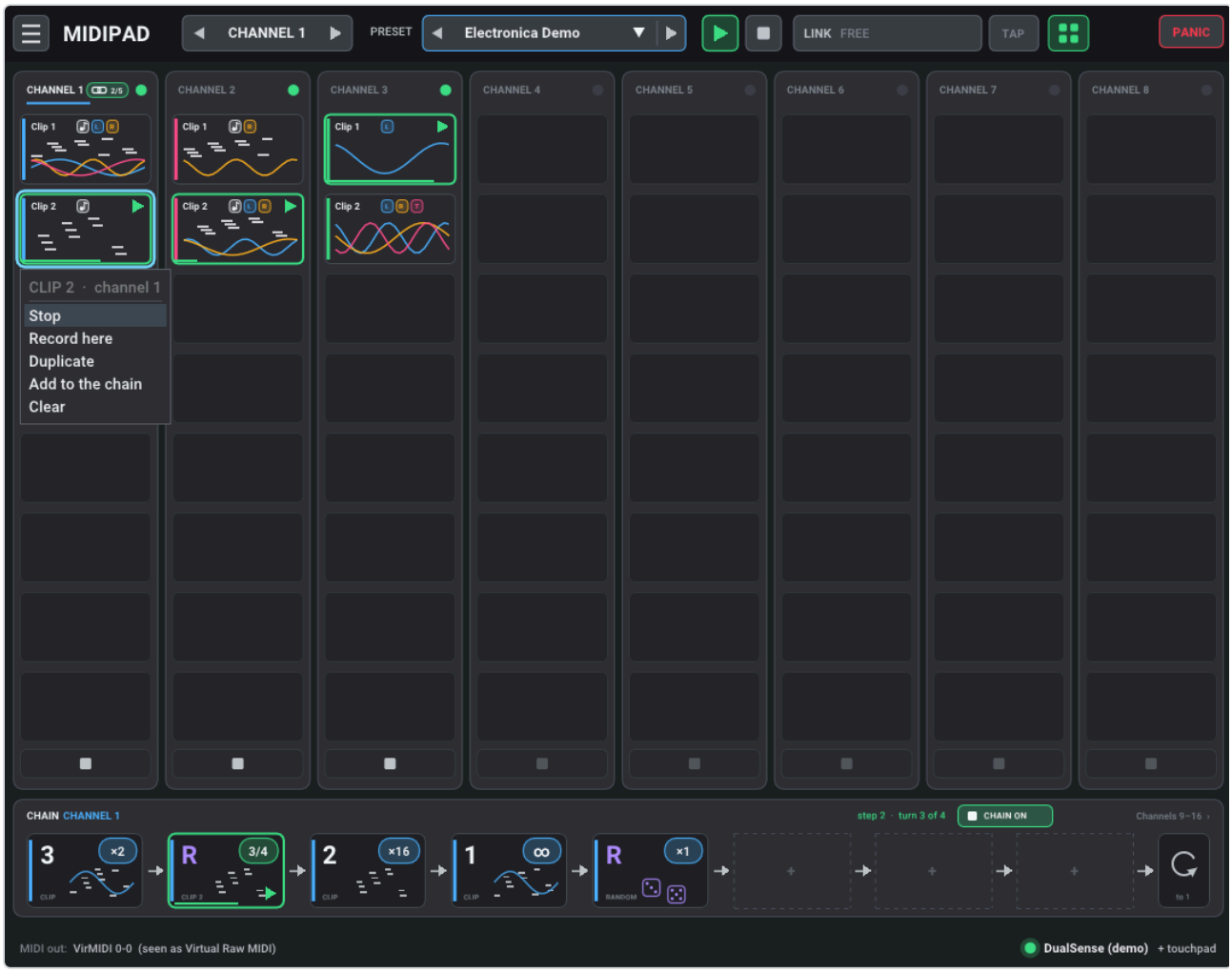
Reading it

- A clip shows its number, small signs for what it holds, in the colours of their modules (a note sign for notes, **L** left stick, **R** right stick, **T** triggers, **P** touchpad), its notes as small bars and its movements as curves.
- The clip playing on a channel has a green outline, a ▶ and a bar along its bottom that shows how far it is in its turn. The one waiting to start has a hollow arrow and blinks; a clip being recorded is outlined in red.

- The cyan frame is the cursor; its channel is the active one. On an empty slot it reads *+ Record: new clip*.

Playing clips

- **Move the cursor** with either stick or the D-pad: a step a push, repeating while held. Left and right change the channel (left of channel 1 comes channel 16, right of 16 comes 1), up and down the slot.
- **Press Select** (L3 or R3 unless you chose another button, chapter 11), or click a clip: it starts at the end of the turn of the clip playing on that channel, from its start. The turn is the note loop's, else the longest movement loop's; with a clock, the clip's length in beats. With nothing playing on the channel, it starts at once.
- **Press Select on the clip playing** (or click it): it stops at the end of its turn, its clips kept. On a clip waiting to start, or while a stop is waiting, the press calls it off. The square at the bottom of a column stops that channel too.
- **Press Menu** (or right-click a clip, a step or a column's header) for its menu: duplicate a clip, clear it, add it to the chain, and more (chapter 4).
- While the session screen shows, the sticks, the D-pad and the Select, Menu and Back buttons send, play and record nothing; the other controls play as usual.



Recording a new clip

Put the cursor on an empty slot (or click it) and record with Loop record as usual: the take goes into a new clip there. The clip playing on that channel stops when the recording starts (it is kept), and the new clip plays as the current one once you let go. While Loop record is held, the sticks, the D-pad, L3 and R3 play and record as usual, so every loop can go into the new clip; once you let go, they move the cursor again (a stick once back at rest). On a slot that already holds a clip, Loop record replaces or overdubs that clip. Leaving the session screen, every channel records into its current clip again.

> **Note:** PANIC clears the current clip of every channel; the other clips stay. Sessions keep every clip, which one is current on each channel, and whether it plays.

Chains of clips

Each channel can play its clips in a sequence: its **chain**, up to eight steps played left to right. A step is a clip and how many turns it plays (×1 to ×64, or ∞); then the next step's clip starts, at the end of the turn, from its start. After the last step the chain goes back to the first, or stops the channel. A turn is the clip's turn, as when you launch a clip.



- **Reading it.** The big number is the step's clip, with a sketch of it; the chip its repeats. The step playing is outlined in green, with its turn (3/4) and a progress bar; a step asked for next blinks. A dashed step is empty: the chain skips it. The column of a channel with a chain shows a chain sign, green with its step (2/5) while it runs.
- **R, a random step** (violet, with two dice): each time the chain comes to it, it picks one of the channel's recorded clips at random and plays it its repeats; while it plays, it names the clip it picked.
- **∞** plays its clip until you ask for the next step (below) or stop the channel.
- **The end cell** shows a round arrow (back to step 1, the default) or a square (the channel stops after the last step). Click it, or press L3 on it, to switch.
- **CHAIN ON / OFF** starts the chain from its first step (or from the step under the cursor) or turns it off; the header then reads *step 2 · turn 3 of 4*. Launching or stopping a clip by hand, recording into another clip and PANIC turn the chain off; its steps stay.

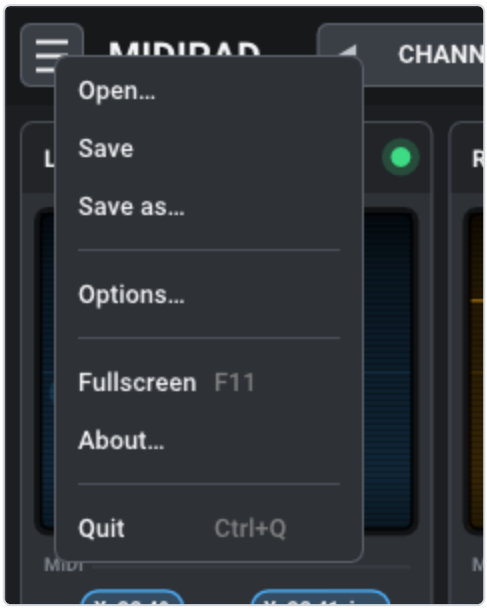
From the controller, move the cursor down past the last clip of a column to enter the chain, and up to leave it. There:

- The left stick or the D-pad goes from step to step.
- The right stick up or down chooses the step's clip (empty, 1 to 8, then R), left or right its repeats (×1 … ×64, then ∞). In the chain the right stick sets values instead of moving the cursor.
- **Select** starts the chain from that step. On the step playing, it goes on to the next step at the end of the turn (this is how an ∞ step ends); on another step, the chain goes there at the end of the turn.
- **Menu** opens the step's menu: Start from here, Random (R), Insert a step, Delete the step, Empty it; on the end of the chain, the channel's. With L3 and R3 as Select (the default), R3 also empties the step.

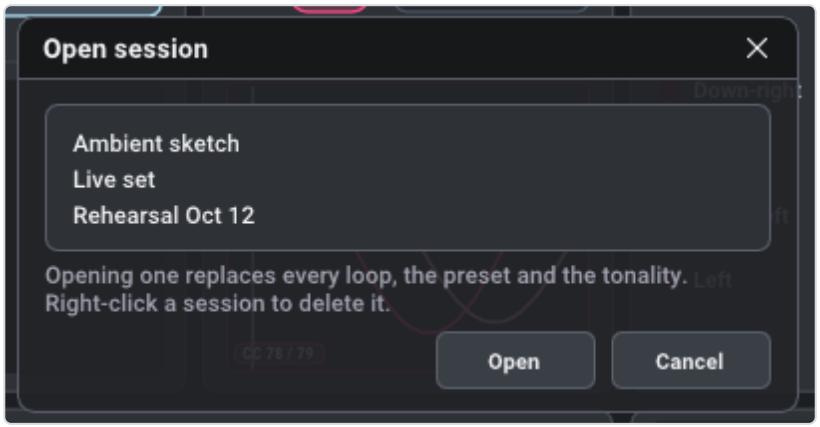
With the mouse: drag a clip from the grid onto a step; turn the wheel over a step's number to choose its clip (past 8: R), over its chip to choose its repeats; right-click a step for its menu. Sessions keep the chains.

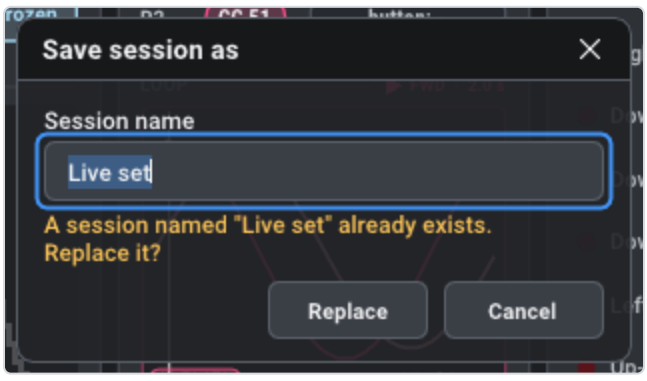
Sessions: saving and opening loops

Loops live in memory: quitting MidiPad loses them, unless you save them in a session. A session keeps every channel's clips and their loops (notes, sticks, triggers, touchpad, with their stretch, amplitude, sync and lock), the chains, a copy of the preset they play with, the tonality, the active channel and the tempo.



- ☰ › **Save as…** names the session and saves it. A name already used asks first: *Replace* overwrites that session.
- ☰ › **Save** saves the current session again; the first time, it asks for a name.
- ☰ › **Open…** lists the saved sessions: double-click one, or select it and click *Open*. Opening replaces every loop, the preset and the tonality; unsaved changes to the preset are asked about first. To delete a session, right-click it, choose *Delete…* and confirm.





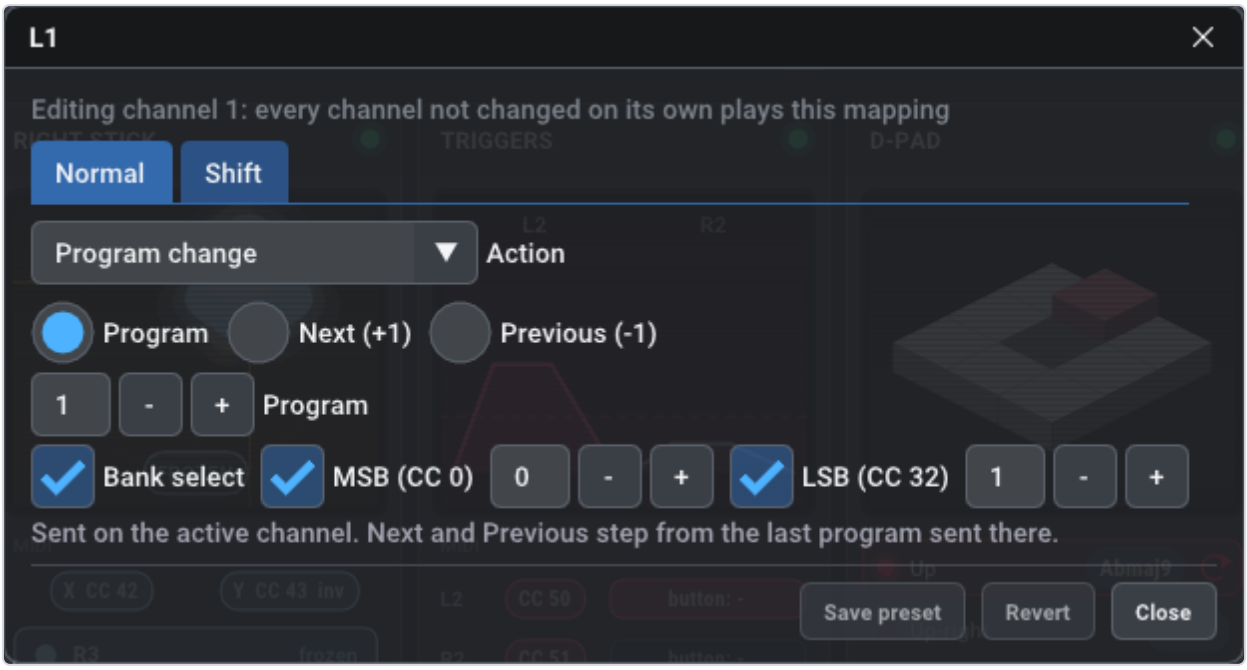
Sessions are files ending in `.midipad`, in `~/.local/share/MidiPad/MidiPad/Sessions/`.

On opening, the preset comes back as it was: the preset of that name if it is unchanged, otherwise the session's copy, marked as changed (save it to keep it). Synced loops find the beat again under whichever clock runs: a session saved with the internal clock opens with Ableton Link, and the other way round. The session's tempo goes to the internal clock, or to the Link session only when MidiPad is alone in it: opening a session never changes another app's tempo.

10. Program Change and transport

MidiPad can change the sound of your synthesizer with MIDI Program Change messages, from a button or by sweeping an axis, and can send MIDI transport commands.

Program Change from a button



Give a button the **Program change** action, then choose:

- **Program:** always selects the same program (1–128).
- **Next (+1)** or **Previous (-1):** steps from the last program sent on the channel. The first press sends program 1. With **Wrap around**, stepping past 128 returns to 1 and the other way round; without it, stepping stops at the ends.
- **Bank select** (optional): sends a bank before the program, with **MSB (CC 0)** and/or **LSB (CC 32)**, each switched on separately.

The message goes out on the active channel. The pad label shows *PC 12* for a fixed program, *PC +1* or *PC -1* for a step.

Program Change from an axis

A stick axis, a trigger or a touchpad finger axis can select a program instead of sending a CC. In its axis window, choose **Program change** instead of CC, then set:

- **From** and **To:** the range of programs the travel covers, for example 1 to 16.
- **Invert:** reverses the direction.

- **Bank select:** a bank of its own for this axis, sent before every program.

Aim, wait, release. The program is sent only once the axis has stayed on it for 150 ms, so sweeping across the range does not load every program on the way. Nothing is sent when the axis springs back to rest, so the last program you chose stays. The axis pill shows *PC 1-16*.

Because each axis has its own bank, two axes can browse two different banks: the left stick one set of sounds, the right stick another.

Numbering

MidiPad shows programs from 1 to 128, as most synthesizers do; *Program 1* sends the value 0. Bank MSB and LSB are shown and sent as raw values (0–127). If your synthesizer counts banks from 1, enter the bank number minus 1.

> **Example: Arturia instruments.** Arturia's software instruments select presets inside a playlist: Bank MSB selects the playlist, Bank LSB the song, and the program the preset in that song, all counted from 0. For playlist 1, song 1, set MSB 0 and LSB 0; then *Program 1* loads the first preset of the song.

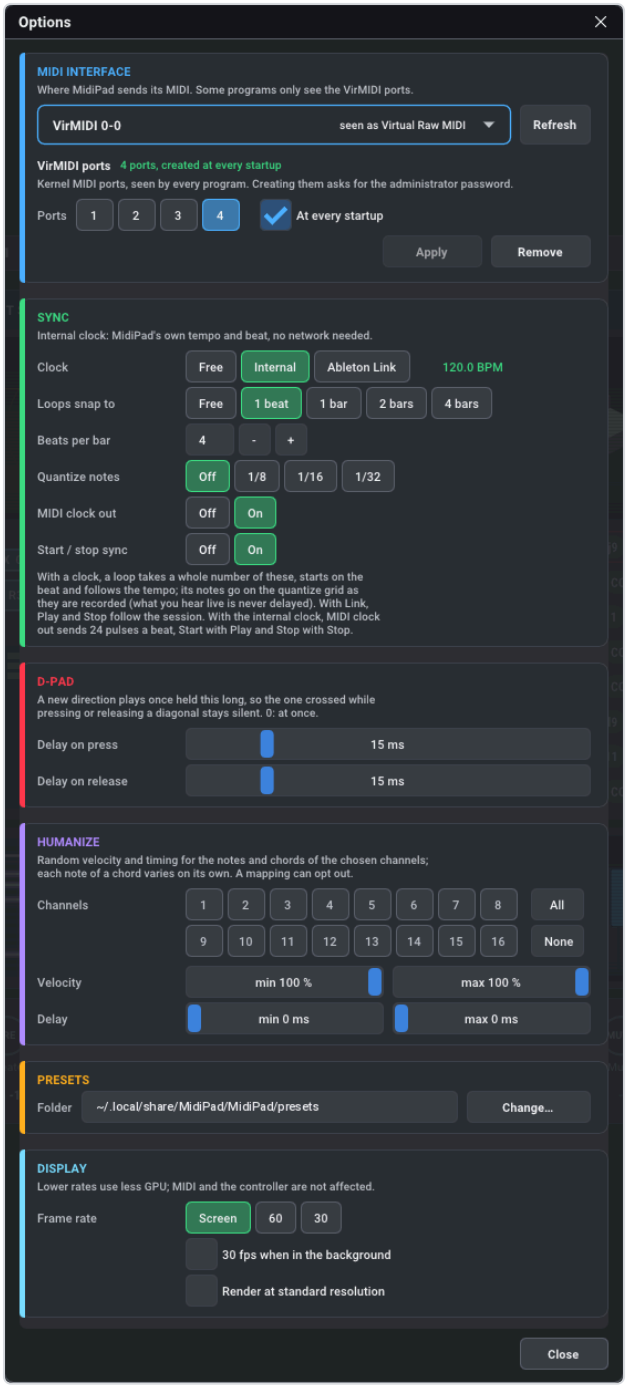
> **Note:** your DAW must pass Program Change messages on to the plugin. Some hosts do not forward them to VST3 plugins; if nothing happens, check with a MIDI monitor placed before the plugin, and try the VST2 version of the plugin.

Transport

The **Transport** action sends MIDI **Start**, **Stop** or **Continue**. These are system real-time messages: your DAW reacts to them only when it is set to follow external MIDI sync. Many DAWs offer MIDI learn on their transport buttons instead; in that case, use a **CC** action on the button and learn it in the DAW.

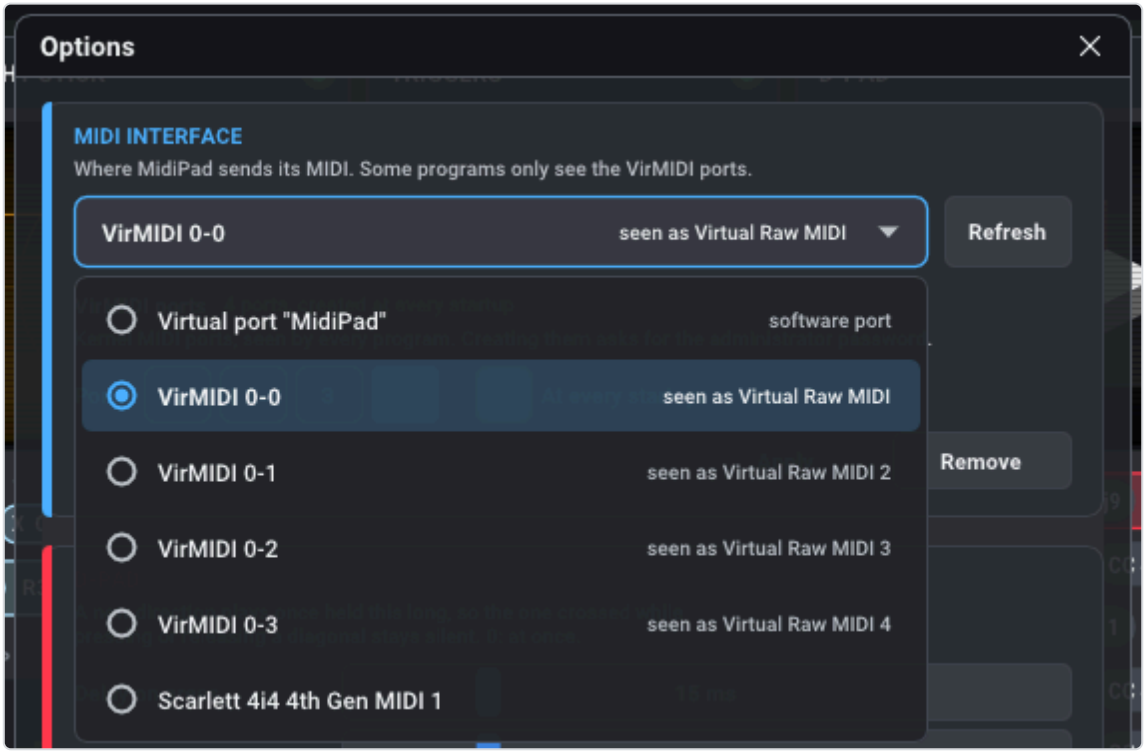
11. Options

The Options window holds the settings that belong to your computer rather than to a preset: the MIDI output, the VirMIDI ports, the clock (internal or Ableton Link), the quantizer, MIDI clock out, the D-pad delay, humanization, the presets folder and the frame rate. Open it from ☰ › **Options**. Settings are saved at once, in `~/.local/share/MidiPad/MidiPad/settings.json`.



MIDI INTERFACE

The drop-down menu shows where MidiPad sends its MIDI. Click it to see every output:



- **Virtual port "MidiPad"** (*software port*): created by MidiPad itself, for programs that read ALSA sequencer ports.
- **VirMIDI N-M** (*seen as Virtual Raw MIDI …*): kernel MIDI ports. The hint on the right is the name under which programs that only list hardware devices show this port.
- **Hardware interfaces**, such as a USB MIDI interface, to drive an external synthesizer.

Refresh updates the list after you plug in a device. If the saved output is missing at startup, the menu shows it in red, marked *(not found)*, and MidiPad sends to its virtual port until you choose another.

> **Note:** the card number N of a VirMIDI port can change between reboots, depending on the order in which devices are found. MidiPad finds its port again regardless.

VirMIDI ports

Some programs only see the kernel's raw MIDI devices and ignore software ports such as *MidiPad*. The `snd-virmidi` kernel module provides such devices, and MidiPad can load it for you:

1. Under **Ports**, choose how many ports you want, from 1 to 4.
2. Tick **At every startup** to create them again after each reboot.
3. Click **Apply**. Your system asks for the administrator password, in its own dialog.

The status line shows how many ports exist and whether they come back at startup. **Remove** unloads the module and deletes the startup files; close the programs using the ports first.

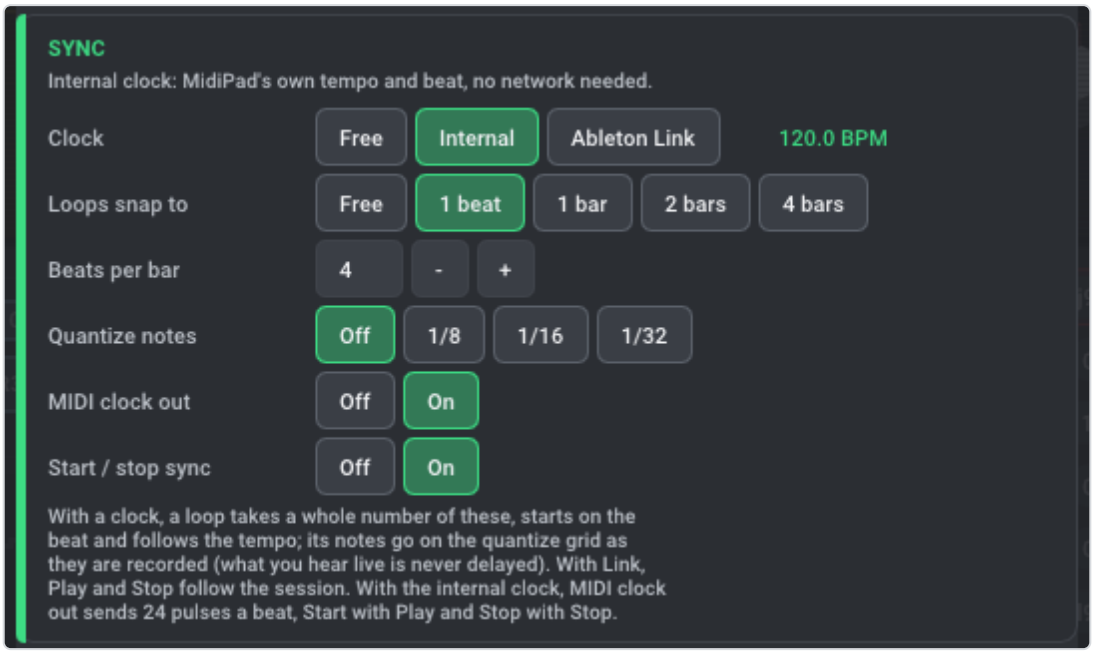
With *At every startup*, MidiPad writes two small files: `/etc/modules-load.d/midipad-virmidi.conf` and `/etc/modprobe.d/midipad-virmidi.conf`. To load the module by hand instead:

```
sudo modprobe snd-virmidi midi_devs=4
```

Then choose **VirMIDI N-0** in MidiPad and, in the receiving program, the input *Virtual Raw MIDI* (VirMIDI N-1 appears as *Virtual Raw MIDI 2*, and so on).

SYNC

How MidiPad keeps time: no beat (free), its own internal clock, or Ableton Link. See *Playing in time* in the Loops chapter.



- **Clock**: *Free* (the default), *Internal* or *Ableton Link*, as the clock button of the top bar. The tempo shows beside it; with Link, the number of peers too.
- **Loops snap to**: the unit a loop recorded with a clock is rounded to: *1 beat* (the default), *1 bar*, *2 bars* or *4 bars*. *Free* keeps loops free even with a clock.
- **Beats per bar**: 1 to 16 (4 by default), for the bars shown in the headers, the grid and the clock button.
- **Quantize notes**: *Off* (the default), *1/8*, *1/16* or *1/32*: the grid the notes recorded into a synced note loop go on (see the Loops chapter).
- **MIDI clock out** (with the internal clock): On (the default) sends MIDI Clock, 24 pulses a beat, with Start on ▶ and Stop on ■, so external devices follow MidiPad's tempo; Off sends none.

- **Start / stop sync** (with Ableton Link): *On* (the default): ▶ and ■ start and stop the Link session, and the loops follow when another app does. *Off*: they only act on MidiPad.

D-PAD

Delay on press and **Delay on release** (0–80 ms, 15 ms by default) set how long a new D-pad direction must hold before it plays. They keep the arrow crossed while pressing or releasing a diagonal from sounding. Set 0 to play every direction at once.

HUMANIZE

Humanization varies velocity and timing at random, for a less mechanical feel.

- **Channels:** the channels it applies to. **All** and **None** select every channel or none.
- **Velocity:** a range in percent of the velocity set in the mapping (100 % by default).
- **Delay:** a range of delay, from 0 to 500 ms.

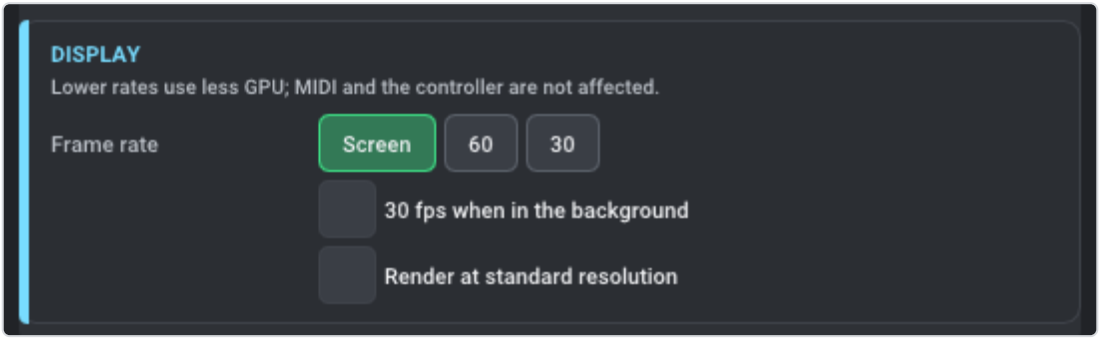
Every note, including each note of a chord, draws its own velocity and delay within these ranges. Its release is delayed by the same amount, so its length does not change. A control can opt out with the **Humanize** box of its mapping window. PANIC also drops notes still waiting to play.

PRESETS

Folder shows the folder MidiPad reads presets from; **Change…** picks another one. The default is `~/.local/share/MidiPad/MidiPad/presets`.

DISPLAY

How many frames a second the window draws. The view moves all the time, so it is drawn again on every frame; on a fast screen (120, 144 or 240 Hz) and a small graphics chip, that can keep the GPU busy.



- **Frame rate**: Screen (the default) draws as often as the screen refreshes; 60 or 30 draw at most that many frames a second, for less GPU work. The view stays smooth at 60.
- **30 fps when in the background**: while another window has the focus (your DAW, say), the window draws at most 30 frames a second, and goes back to its frame rate as soon as you click it. Off by

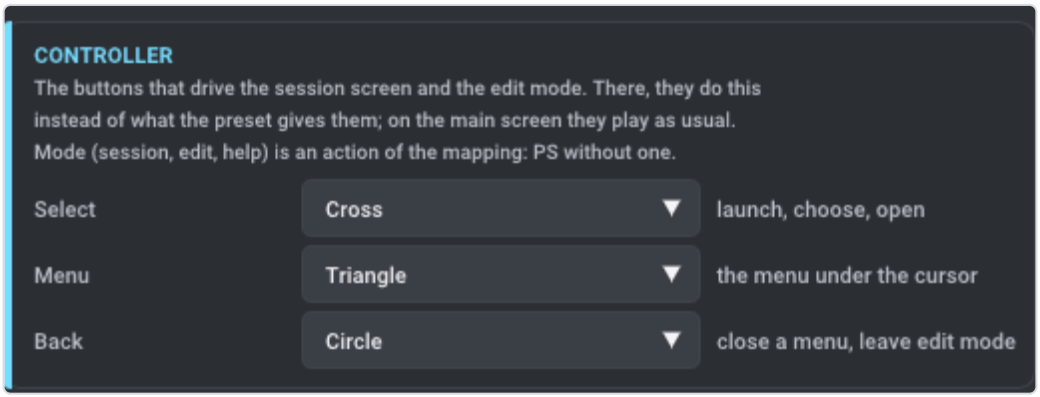
default.

Render at standard resolution (off by default, from the next start): on a scaled screen (125 %, 200 % on Wayland), MidiPad draws its window at its size in points and the desktop stretches it to the screen, instead of drawing every pixel of the screen itself. At 125 %, that is 36 % fewer pixels to draw, for a slightly less sharp picture. Where the screen is not scaled, or with X11, it changes nothing, and the card says so.

MIDI and the controller do not depend on it: they run on their own thread, 1,000 times a second.

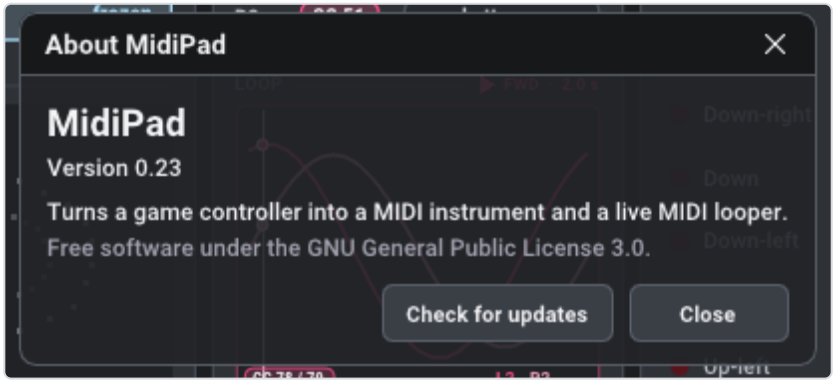
CONTROLLER

The buttons that drive the session screen and the edit mode (chapter 4): **Select** (L3 / R3 by default), **Menu** and **Back** (none by default). Choose, for example, Cross, Triangle and Circle. There, these buttons do this instead of what the preset gives them; on the main screen they play as usual, so a Freeze on L3 and R3 keeps working once Select is on another button. The **Mode** button itself is an action of the mapping (chapter 6). These settings belong to MidiPad, not to a preset.



About and updates

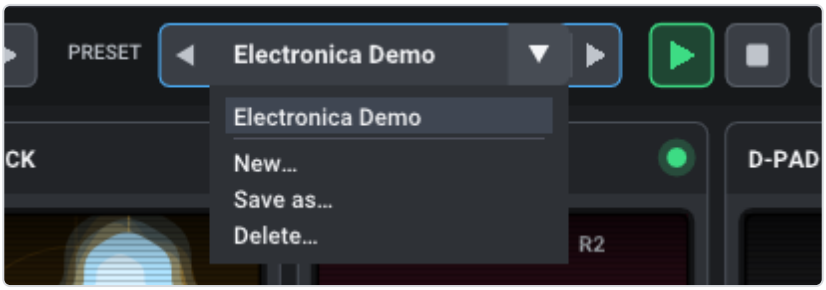
☰ › **About…** shows MidiPad's version and license. **Check for updates** asks GitHub whether a newer version exists, only when you click (through `curl`): *MidiPad 0.24 is available.* comes with a **Download page** button, *You have the latest version.* otherwise. If GitHub cannot be reached, the window says so and offers the releases page.



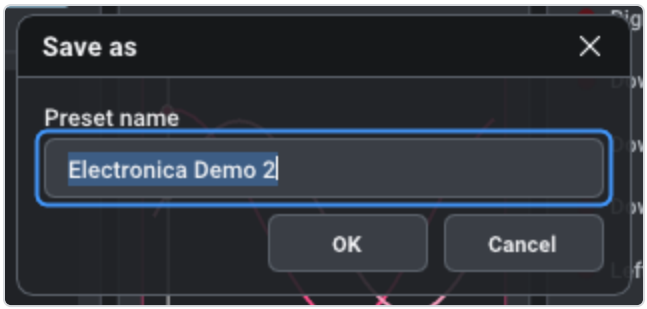
12. Presets

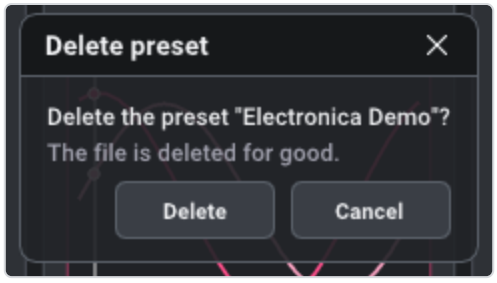
A preset is a complete layout: channel 1's mapping, the other channels' own mappings and the loop settings, stored in one JSON file. Switch presets to move from one song or setup to another.

The PRESET menu



- Click the preset name to open the list: every preset of the presets folder, then **New...**, **Save as...** and **Delete...**.
- The ◀ and ▶ arrows on each side load the previous or next preset.
- **New...** creates an empty preset, asking for its name.
- **Save as...** saves a copy of the current preset under a new name and switches to it.
- **Delete...** deletes the current preset after a confirmation. The file is deleted for good.





Saving

Every change is heard at once but kept in memory until you save. Press `Ctrl+S` or click **Save preset** in any mapping window. If you open another preset or quit with unsaved changes, MidiPad asks whether

to **Save**, **Don't save** or **Cancel**.

MidiPad reopens the last preset used at the next start. To start from a given file instead, pass it on the command line; its folder becomes the presets folder:

```
midipad ~/Music/MidiPad/live-set.json
```

The preset file

Presets are plain text files, `<name>.json`, in the presets folder. You can read, copy, share and edit them in any text editor; after editing a preset that is open in MidiPad, load another preset and come back to it with the arrows to load your changes, rather than clicking **Revert**, which only discards changes made in the window. Here is a short preset: Cross plays a chord, the left stick sends two CCs, and channel 16 has its own mapping.

```
{
    "format": "midipad-preset",
    "version": 2,
    "name": "My Preset",
    "base": {
        "buttons": {
            "south": { "type": "chord", "chord": "Fm9", "octave": 4, "root_below": 2
},
            "right_shoulder": { "type": "loop_record" }
        },
        "dpad": {
            "up": { "type": "chord", "chord": "Abmaj9", "mode": "toggle" }
        },
        "axes": {
            "left_x": { "cc": 40 },
            "left_y": { "cc": 41, "invert": true }
        }
    },
    "channels": {
        "16": {
            "buttons": { "south": { "type": "note", "note": 53 } }
        }
    }
}
```

The top level holds `base` (channel 1's mapping, which every channel without its own plays) and, optionally, `channels`: one mapping per channel number, from "2" to "16" (a mapping under "1", from older files, is read as the base one). Each mapping has up to six sections: `buttons`, `dpad`, `axes`, `touchpad`, `loops` and `shift` (its Shift layer). Fields left out take their default value. Chapter 16 lists every key.

> **Note:** if a preset file contains an error, MidiPad refuses it and names the faulty entry, for example `base.buttons.south.note: expected a MIDI note number or a name like "C4"`.

13. Using MidiPad with a DAW on Linux

MidiPad works with any DAW or synthesizer that receives MIDI. The only decision is which port to use, and it depends on how your DAW lists MIDI inputs.

Choosing the port

Your DAW lists...	Output to choose in MidiPad	Input to choose in the DAW
ALSA sequencer ports (most Linux DAWs)	Virtual port "MidiPad"	*MidiPad*
Only hardware MIDI devices	VirMIDI N-0	*Virtual Raw MIDI* (or the name shown next to the port in MidiPad)
JACK MIDI ports only	Virtual port "MidiPad", bridged with `a2jmidid`	*MidiPad* under the a2j bridge
An external synthesizer	Your USB MIDI interface	—

When in doubt, use a VirMIDI port: every program sees it.

Assigning CCs with MIDI learn

Most DAWs and synthesizers assign a CC to a parameter by MIDI learn: you click the parameter, then move the control. Sticks send two CCs at once, which can confuse MIDI learn. To learn one axis alone:

1. Open the stick window and untick **Enabled** for the axis you do not want to learn.
2. Start MIDI learn in the DAW and move the stick.
3. Tick **Enabled** again.

The same applies to loops: their CCs (70–79 by default) are separate from the live CCs, so learn them separately. Record a short loop, then start MIDI learn while it plays.

> **Tip:** use **Axis freeze** to hold a value steady while you assign or check a parameter.

A multitimbral setup

1. Create one instrument track per sound, each listening to the MidiPad port on its own MIDI channel.
2. In MidiPad, give each channel the layout it needs (chapter 8).
3. Put **Channel change** on two buttons to move between instruments while playing.
4. Record loops on each channel: they keep playing while you move on to the next instrument.

Latency

MidiPad reads the controller 1,000 times per second on a dedicated thread, with real-time priority when the system allows it, and sends MIDI directly from that thread. Drawing the window never delays the MIDI. On a typical machine, processing takes about 20 µs per reading. For the lowest latency, prefer a USB connection to Bluetooth.

> **Note:** for real-time priority, your user must be allowed to use it, usually by being in the `audio` group with the limits most audio distributions set up.

14. Running without a window

MidiPad can play a preset without its window, from a terminal: useful on a small computer dedicated to the stage, or to start MidiPad from a script. It reads the first controller connected, before or after the start, and creates a virtual MIDI port named *MidiPad* unless you choose another output.

```
midipad --headless ~/.local/share/MidiPad/MidiPad/presets/"Electronica Demo.json"
```

The preset reloads automatically whenever its file is saved, so you can edit it in a text editor and hear the result at once. If the new version contains an error, the previous one stays active and the error is printed.

Option	Effect
`-c`, `--connect NAME`	Send to an existing port whose name contains NAME, instead of creating a virtual port (example: `-c "VirMIDI 0-0"`)
`-n`, `--port-name NAME`	Name of the virtual port (default: MidiPad)
`--api alsa|jack`	MIDI system to use on Linux (default: alsa)
`--channel N`	Starting MIDI channel, 1 to 16 (default: 1)
`-m`, `--monitor`	Print every MIDI message sent
`-i`, `--input`	Print the buttons pressed
`--stats`	Print the measured processing times on exit
`--poll-hz N`	Controller readings per second (default: 1000)
`--list-ports`	List the MIDI output ports and exit
`-h`, `--help`	Show the help

Press `Ctrl+C` to stop. Loops and all other features of the preset work as in the window; the Options window's settings do not apply: the D-pad delay stays at 15 ms and humanization is off.

15. Troubleshooting

Most problems come from the route between MidiPad and the receiving program. Check that route first: MidiPad's status bar shows where it sends, and the command `aseqdump` shows what it actually sends.

Checking what MidiPad sends

Switch MidiPad's output to the virtual port "MidiPad" for the test, then watch it while you play:

```
aseqdump -l
```

```
aseqdump -p MidiPad
```

Every press and movement prints a line: *Note on*, *Control change*, *Program change* and so on, with its channel. If lines appear, MidiPad works and the problem lies in the route or the receiving program. The first command lists every port; a VirMIDI port cannot be watched this way, because it shows what leaves the port on the other side, not what MidiPad sends into it.

Common problems

Symptom	Cause	Solution
The status bar shows *No gamepad*	Controller not connected or not paired	Reconnect it by USB, or pair it again over Bluetooth (chapter 2)
+ touchpad is missing, the TOUCHPAD module stays still	No access to the touchpad device	Install the udev rule (chapter 2) and reconnect the controller
The DAW does not list MidiPad	The DAW only lists hardware devices	Use a VirMIDI port (chapter 11)
The MIDI output window appears at every start	The saved port is missing	Choose a port again; with VirMIDI, tick **At every startup**
No sound, but `aseqdump` shows messages	The track does not listen to the port or channel	Set the track's input to the port, on all channels; arm or monitor the track
A stick moves two parameters during MIDI learn	Both axes send at once	Untick **Enabled** for the other axis while learning (chapter 13)
A parameter jumps between two values	A loop and a live axis use the same CC	Give the loop its own CCs (red warning in the window)

Symptom	Cause	Solution
The D-pad plays a wrong direction on diagonals	Delay too short for your playing	Raise the D-pad delay in Options
Program changes have no effect	The host does not pass them to the plugin, or the numbering differs	Check with a MIDI monitor before the plugin; try the VST2 version; check the bank numbering (chapter 10)
Start / Stop do nothing in the DAW	The DAW does not follow external MIDI sync	Enable external sync, or use a CC with MIDI learn (chapter 10)
Notes keep sounding	An instrument missed a note-off	Click **PANIC**
VirMIDI: *The VirMIDI ports are in use*	A program still has the ports open	Close it, then **Apply** or **Remove** again
VirMIDI: *Not authorized*	The password was refused or the dialog closed	Click **Apply** again and enter the administrator password

Reporting a problem

When something behaves unexpectedly, note the MidiPad version, the controller and how it is connected (USB or Bluetooth), the output port, and the output of `aseqdump` while you reproduce the problem.

16. Reference

Keyboard shortcuts

Shortcut	Action
`Ctrl+S`	Save the preset
`Ctrl+Q`	Quit (asks about unsaved changes)
`F11`	Fullscreen on or off
`Esc`	Close the window or dialog in front
`Wheel, double-click` on the tempo	Change the clock's tempo, or type it (chapter 9)
`Drag` in the note loop editor	Move loop notes, select them in a rectangle, change their length by their end, or the loop's by the right edge (chapter 9)
Right-click on a note in the editor, or on a LOOP panel or the piano roll	Delete the note, or the selection it belongs to (the Delete key too); elsewhere (a lane, a LOOP panel, the piano roll, a clip, a chain step, a channel), its menu (chapter 4)

Files and folders

Item	Location
Program	`build/release/src/app/midipad` after building
Settings	`~/.local/share/MidiPad/MidiPad/settings.json`
Presets (default)	`~/.local/share/MidiPad/MidiPad/presets/`
Sessions	`~/.local/share/MidiPad/MidiPad/Sessions/` (.midipad files)
Touchpad access rule	`/etc/udev/rules.d/70-dualsense.rules`
VirMIDI at startup	`/etc/modules-load.d/midipad-virmidi.conf`, `/etc/modprobe.d/midipad-virmidi.conf`

Control names

The names used in preset files, with the DualSense labels shown in the window:

Preset key	DualSense	Section
`south`, `east`, `west`, `north`	Cross, Circle, Square, Triangle	`buttons`
`left_shoulder`, `right_shoulder`	L1, R1	`buttons`
`left_trigger`, `right_trigger`	L2, R2 (as buttons)	`buttons`
`left_stick`, `right_stick`	L3, R3 (stick clicks)	`buttons`
`back`, `start`, `guide`	Create, Options, PS	`buttons`
`touchpad`, `misc1`	Touchpad click, Mute	`buttons`
`up`, `up_right`, `right`, `down_right`, `down`, `down_left`, `left`, `up_left`	D-pad directions	`dpad`
`left_x`, `left_y`, `right_x`, `right_y`	Stick axes	`axes`
`left_trigger`, `right_trigger`	L2, R2 (as axes)	`axes`
`finger1`, `finger2`	Touchpad fingers	`touchpad`
`top_left`, `top_right`, `bottom_left`, `bottom_right`	Touchpad zones	`touchpad.fingerN.zones`
`left`, `right`, `finger1`, `finger2`, `triggers`, `notes`	Loops	`loops`

Action fields

`type`	**Fields (default)**
`note`	`note` (number or name such as "C4"), `velocity` (100), `mode` ("momentary" or "toggle"), `humanize` (true)
`cc`	`cc`, `value` (127), `off_value` (0), `mode`
`chord`	`chord` (name), `octave` (4), `root_below` (0, 1 or 2), `velocity`, `mode`, `humanize`
`transport`	`command`: "start", "stop" or "continue"
`channel`	`step`: 1 or -1
`freeze`	`axes`: list of axis names
`loop_record`	—
`program`	`program` (0–127) or `step` (1 or -1) with `wrap` (true); `bank_msb`, `bank_lsb` (optional)
`shift`	`mode` ("momentary" or "toggle"); `loop_stretch`, from older files, is read as shift
`tap_tempo`	—

type	Fields (default)
session	—

Axis fields

Field	Meaning (default)
cc	Controller number; required unless program is true
min, max	Values at the ends of the travel (0, 127); programs 0–127 in program mode
invert	Reverse the travel (false)
program	Select programs instead of sending a CC (false)
bank_msb, bank_lsb	Bank sent before each program, in program mode (none)
velocity	true: a Velocity axis instead of a CC (false)
transpose	A Transpose axis: {"down": 12, "up": 12, "to": "notes"}, down and up from 0 to 24 semitones, "to": "notes" or "loop"

A touchpad finger takes mode: "axis" (with x and y, each an axis entry) or "zones" (with zones, each an action). A loop entry takes playback ("forward", "reverse" or "back_and_forth"), cc_x, cc_y and, for an axis in steps, steps_x or steps_y: {"levels": 8, "step": "1/16"} (levels from 2 to 128; step "1/32", "1/16", "1/8", "1/4", "1/2" or "1 bar"). The shift section holds the Shift layer: buttons, dpad and touchpad (finger1 and finger2, each with zones), with the same action entries as the mapping's own; a control left out plays its normal action under Shift too.

Timing values

Value	Setting
Controller readings	1,000 per second
D-pad delay	15 ms on press and release, 0–80 ms in Options
Trigger as button	Press at 25 %, release at 15 % of the travel
Program Change from an axis	Sent after 150 ms on the same program
Overdub tap	Shorter than 500 ms; with Shift, two taps within 350 ms lock or unlock the note loop
Loop recording	200 samples per second, 60 s at most
Note loop	Up to 4,096 note events per channel
Humanize delay	0–500 ms

Value	Setting
Loop length (stretch)	-20x to +100x, steps of 0.5; Shift button + X (or Shift key + wheel)
Loop amplitude	0 % to 800 %, steps of 5 % up to 100 % and of 25 % above; softly clipped at the ends of the CC range; eased in; Shift button + Y (or Ctrl + wheel)
Shift with a stick	Dead zone 25 % of the half travel; 2 to 40 length steps, 2 to 20 amplitude steps a second; 0.4 s pause at 1x and at 100 %
Shift with a finger	One step every 3 % of the touchpad's width or height
Loop sync (Ableton Link)	Length rounded to whole snaps (1 beat by default), one at least; start moved to the nearest snap; 4 beats per bar by default (1–16)
Tempo	20–999 BPM; wheel: 1 BPM a notch, 0.1 with the Shift key; internal clock at 120 BPM at first; MIDI clock out at 24 pulses a beat
Tap tempo	Median of the last 4 intervals; starts over after a 2 s pause
Quantize grid	1/8, 1/16 or 1/32 notes: 2, 4 or 8 steps a beat
Transpose	Up to 24 semitones each way; with a tonality, 12 semitones = one octave of degrees